DG INTERNS HUB
Learn. Build. Grow.

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Internship Track:	Cyber Security Intern
Submission Date:	23 October
Week :	1
Task 1:	Lab Setup,Linux and Network Foundation
Student Id:	DG/15/OCT/CYBER/118

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Table of Contents

1. EXECUTIVE SUMMARY 5

2. INTRODUCTION 5

3. OBJECTIVES 6

4. ACTIVITIES AND METHODOLOGY 6

4.1 Cybersecurity Fundamentals 6

4.2 Networking Foundations 7

4.3 Linux Administration 7

4.4 Virtual Laboratory Setup 8

4.5 Linux Practical Exercises 14

4.6 Networking Practical Exercises 19

4.7 TryHackMe Learning Activities 19

4.8 OverTheWire Bandit 24

5. FINDINGS AND LESSONS LEARNED 35

Key Learning Outcomes 35

6. CHALLENGES AND RESOLUTIONS 36

7. EVIDENCE AND DELIVERABLES 36

8. PROFESSIONAL AND ETHICAL CONSIDERATIONS 37

9. CONCLUSION 37

10. REFERENCES 38

Figure 1 9
Figure 2 9
Figure 3 10
Figure 4 10
Figure 5 11
Figure 6 11
Figure 7 12
Figure 8 12
Figure 9 13
Figure 10 13
Figure 11 14
Figure 12 15
Figure 13 15
Figure 14 16
Figure 15 16
Figure 16 17
Figure 17 17
Figure 18 18
Figure 19 18
Figure 20 20
Figure 21 20
Figure 22 21
Figure 23 21
Figure 24 22
Figure 25 22
Figure 26 23
Figure 27 23
Figure 28 24
Figure 29 25
Figure 30 25
Figure 31 25
Figure 32 26
Figure 33 26
Figure 34 27
Figure 35 27
Figure 36 28
Figure 37 28
Figure 38 29
Figure 39 29
Figure 40 30

Figure 41 30
Figure 42 31
Figure 43 31
Figure 44 32
Figure 45 32
Figure 46 33
Figure 47 33
Figure 48 34
Figure 49 34

1. EXECUTIVE SUMMARY

During Week 1 of my Cyber Security Internship at DG Interns Hub, I focused on learning the basic concepts and technical skills needed for cybersecurity work. The main areas covered were cybersecurity fundamentals, Linux administration, networking, virtual laboratory setup, and practical learning through cybersecurity platforms.

I studied the CIA triad, different cybersecurity roles, networking models, common network services, Linux file permissions, and the importance of working in a safe laboratory environment.

The practical exercises introduced me to Linux commands, network configuration, connectivity testing, and guided cybersecurity challenges on TryHackMe and OverTheWire Bandit.

This report summarizes the activities, lessons learned, and practical work carried out during the first week. The supporting screenshots and practical evidence are stored separately in the Findings folder.

2. INTRODUCTION

Cybersecurity involves protecting computer systems, networks, applications, and information from unauthorized access, attacks, and disruption.

As a Cyber Security Intern, I need to understand how computer systems work, how they communicate, and how security measures help protect them. Linux and networking are important parts of this foundation because many cybersecurity tools and systems depend on them.

Week 1 was focused on developing this foundation through theoretical learning, practical exercises, and documentation.

The activities helped me understand basic security principles, Linux system administration, network communication, and the responsibilities of cybersecurity professionals.

3. OBJECTIVES

The main objectives for Week 1 were:

1. Understand cybersecurity fundamentals and the CIA triad.
2. Learn the differences between SOC analysts, penetration testers, and incident responders.
3. Understand the OSI and TCP/IP networking models.
4. Learn how IP addresses, MAC addresses, DNS, DHCP, and network ports work.
5. Practise Linux administration, user management, groups, and file permissions.
6. Understand virtual machine networking and laboratory isolation.
7. Prepare and document a cybersecurity laboratory environment.
8. Practise Linux and networking commands.
9. Work through the assigned TryHackMe and OverTheWire Bandit activities.
10. Document the activities and prepare evidence of the work completed.

4. ACTIVITIES AND METHODOLOGY

4.1 Cybersecurity Fundamentals

I began by studying the basic principles of cybersecurity and why they are important in protecting digital information.

The topics covered included:

- The meaning and importance of cybersecurity.
- Confidentiality, Integrity, and Availability.
- Threats, vulnerabilities, risks, and exploits.
- The responsibilities of SOC analysts, penetration testers, and incident responders.
- The differences between Blue Team and Red Team activities.
- Legal and ethical responsibilities in cybersecurity.

The CIA triad helped me understand that information should only be accessed by authorized people, remain accurate and reliable, and be available when needed.

I also learned that cybersecurity professionals have different responsibilities. SOC analysts monitor systems and investigate alerts, penetration testers assess security weaknesses with authorization, and incident responders help investigate and manage security incidents.

Learning outcome: I gained a better understanding of cybersecurity principles and the roles played by different cybersecurity professionals.

4.2 Networking Foundations

Networking was another important area of the first week. I studied how computers communicate and how network information can be used to understand system activity.

The topics covered included:

- IPv4 and IPv6 addressing.
- MAC addresses.
- DNS and DHCP.
- TCP and UDP.
- Common network ports and services.
- The OSI networking model.
- The TCP/IP networking model.
- Network communication and connectivity testing.

I also learned about common services such as FTP, SSH, Telnet, DNS, HTTP, HTTPS, SMB, MySQL, and Remote Desktop Protocol.

Understanding these services is important because they are used in everyday network communication. It also helps a security analyst recognize services that may need monitoring or security controls.

Learning outcome: I improved my understanding of network addressing, protocols, services, and how networking knowledge supports security monitoring.

4.3 Linux Administration

During the Linux administration activities, I practised using the command line and learned how Linux systems are organized and managed.

The exercises covered:

- Navigating the Linux filesystem.
- Identifying important directories.
- Creating and managing user accounts.
- Creating groups and managing group membership.
- Understanding file and directory permissions.

- Using chmod and chown.
- Searching for files using find.
- Searching for text using grep.
- Checking running processes using ps.
- Inspecting network sockets using ss.
- Learning how scheduled tasks work through cron.

These exercises helped me become more familiar with basic Linux administration. I also learned why user permissions and system monitoring are important when managing a computer securely.

Learning outcome: I improved my understanding of Linux commands, user and group management, file permissions, process monitoring, and system administration.

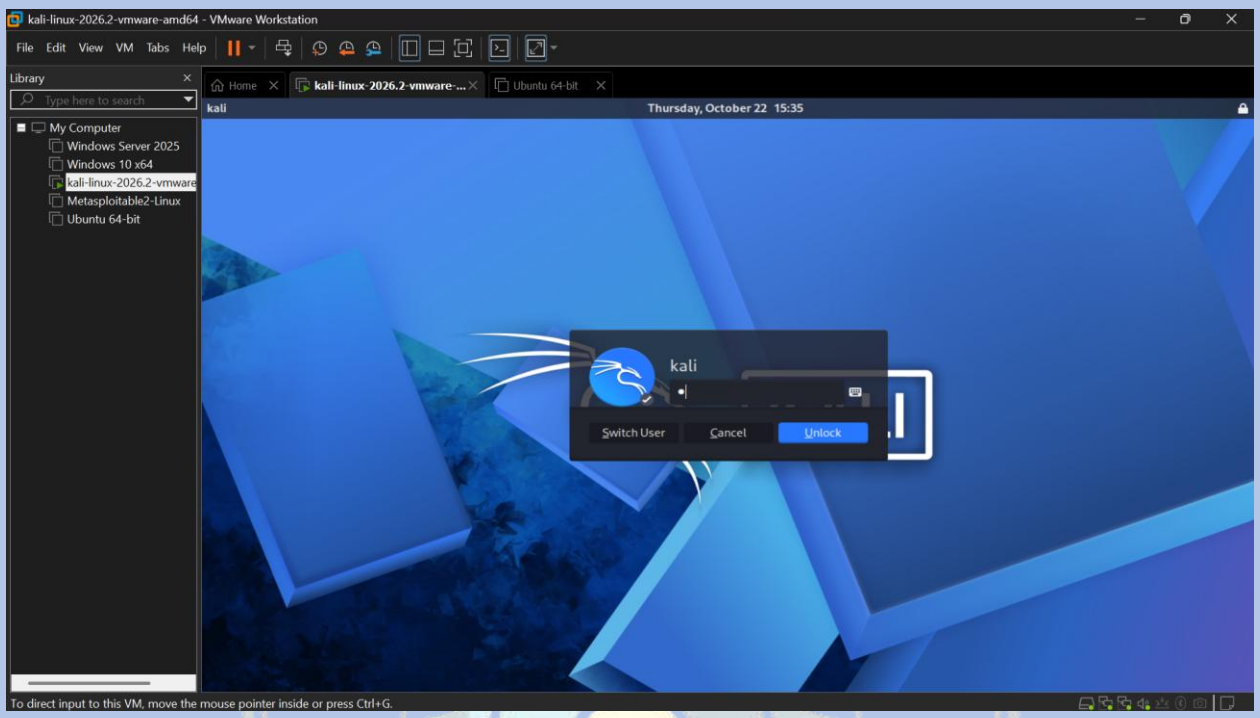
4.4 Virtual Laboratory Setup

The laboratory activity focused on preparing a controlled environment for cybersecurity practice.

The planned laboratory components included:

- Kali Linux virtual machine.
- Ubuntu virtual machine.
- Windows evaluation virtual machine.
- VirtualBox virtualization software.
- Isolated virtual networking.
- Network connectivity testing.
- Virtual machine snapshots.

Figure 1



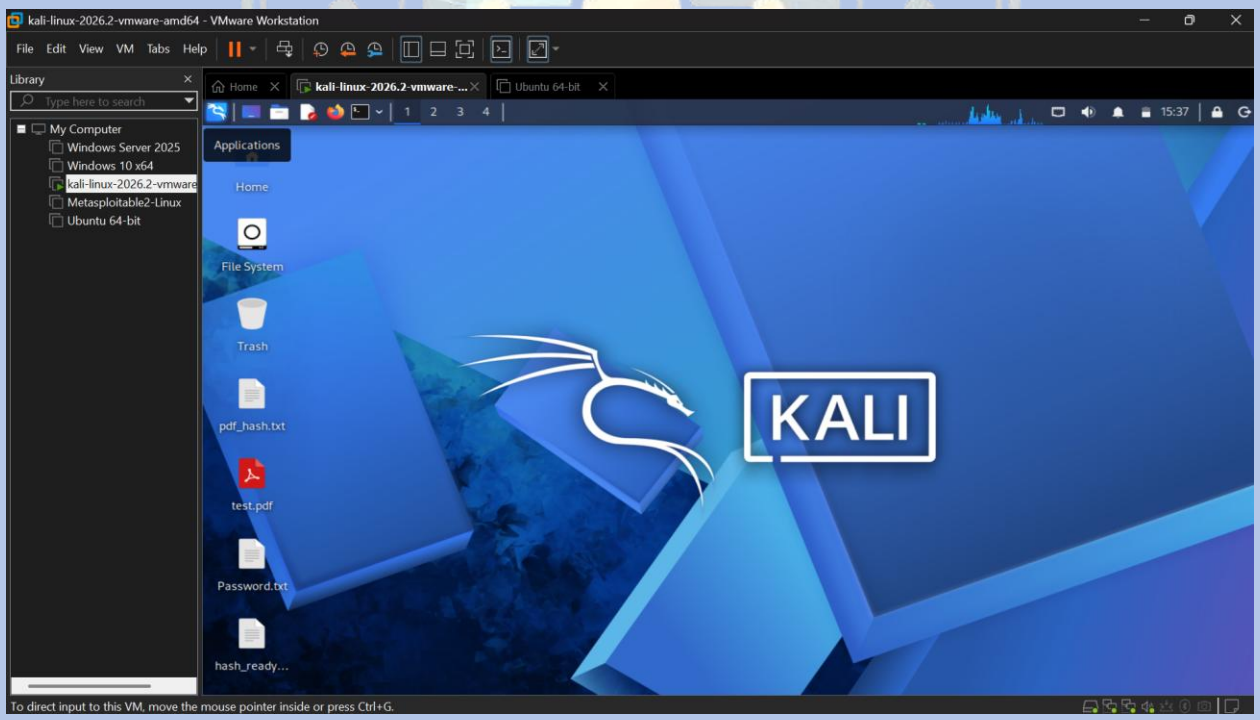


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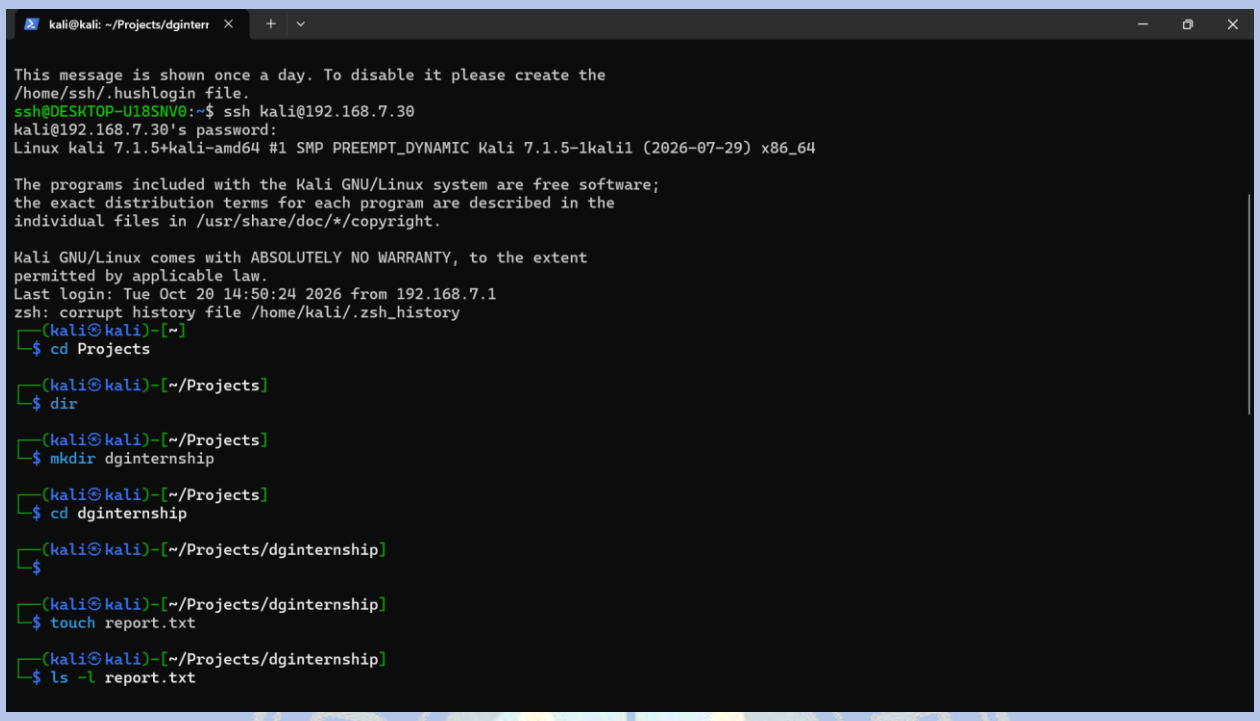


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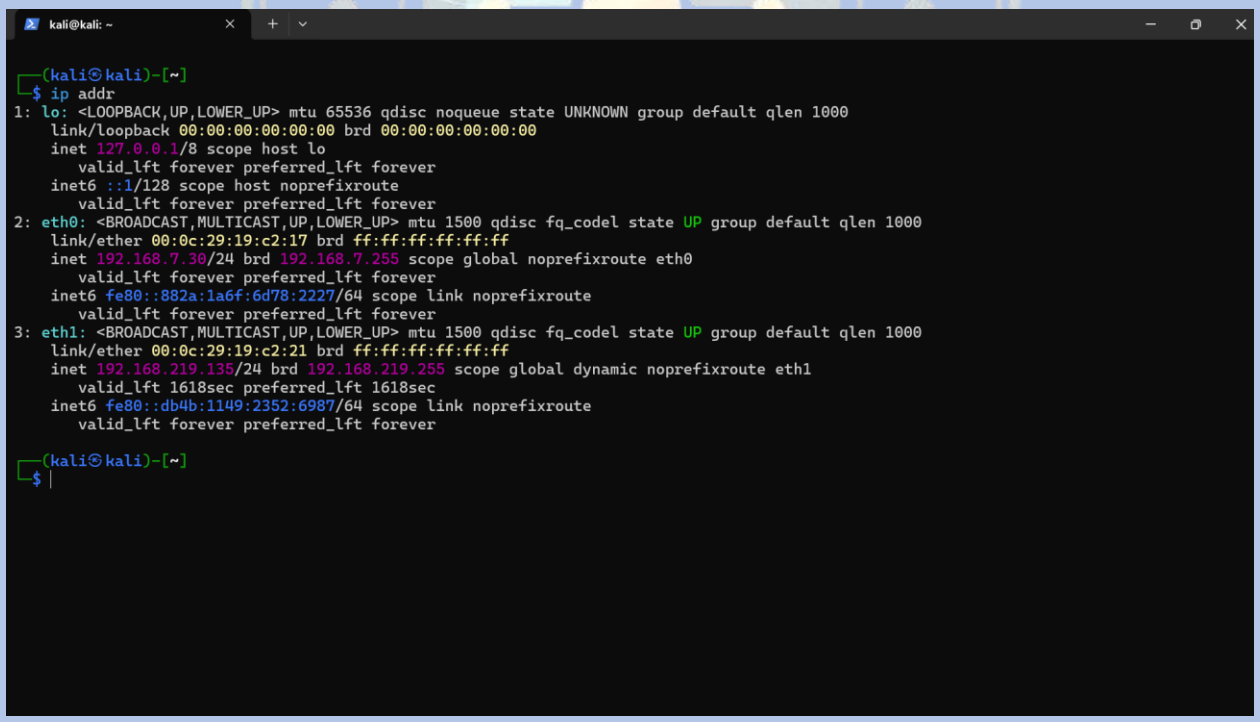


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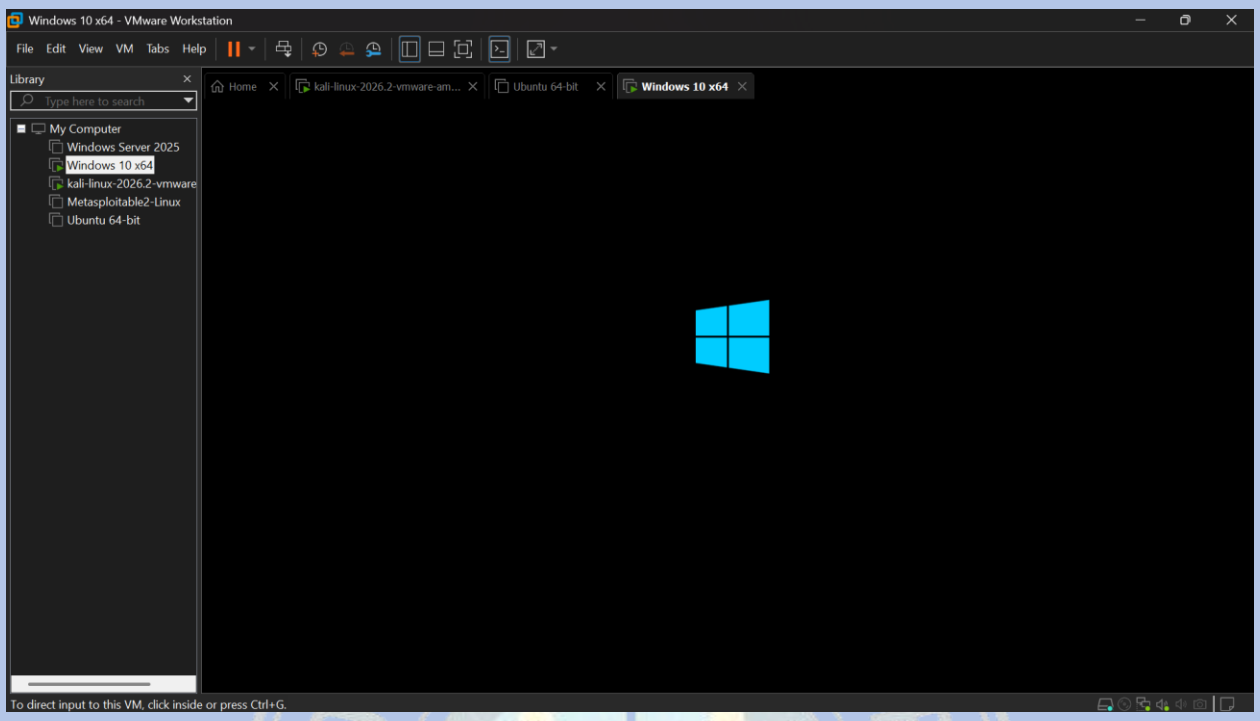


Figure 5

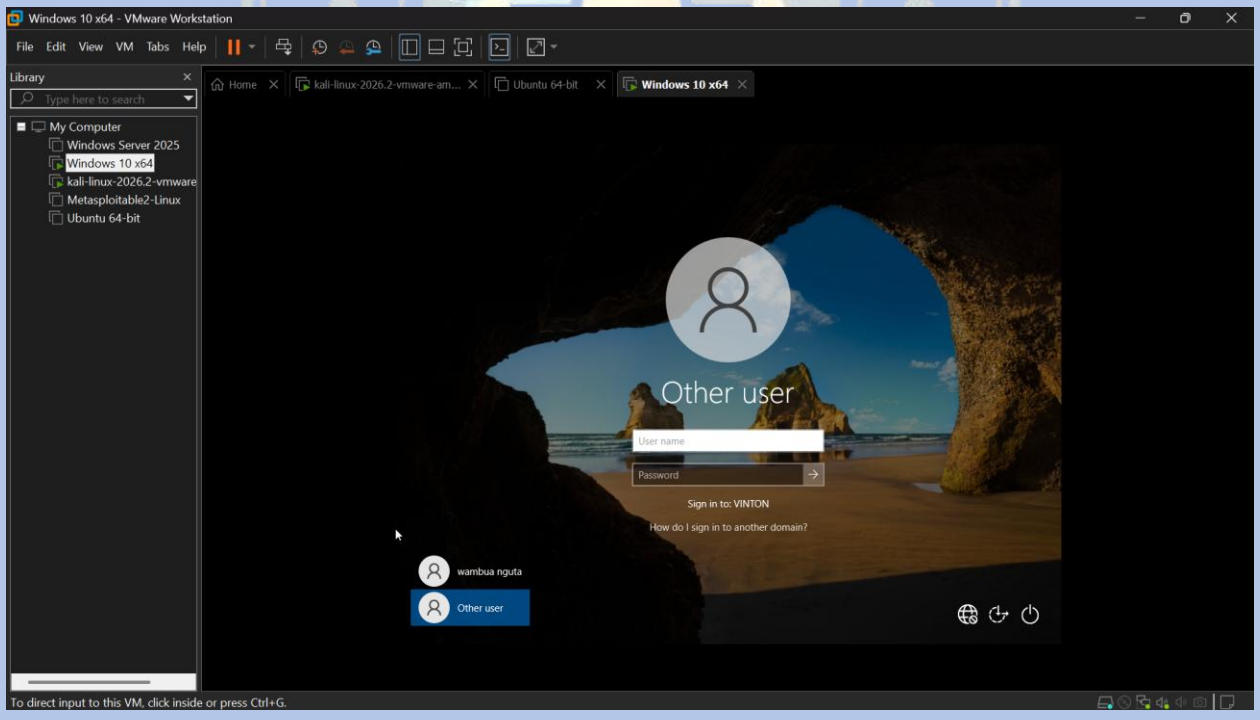


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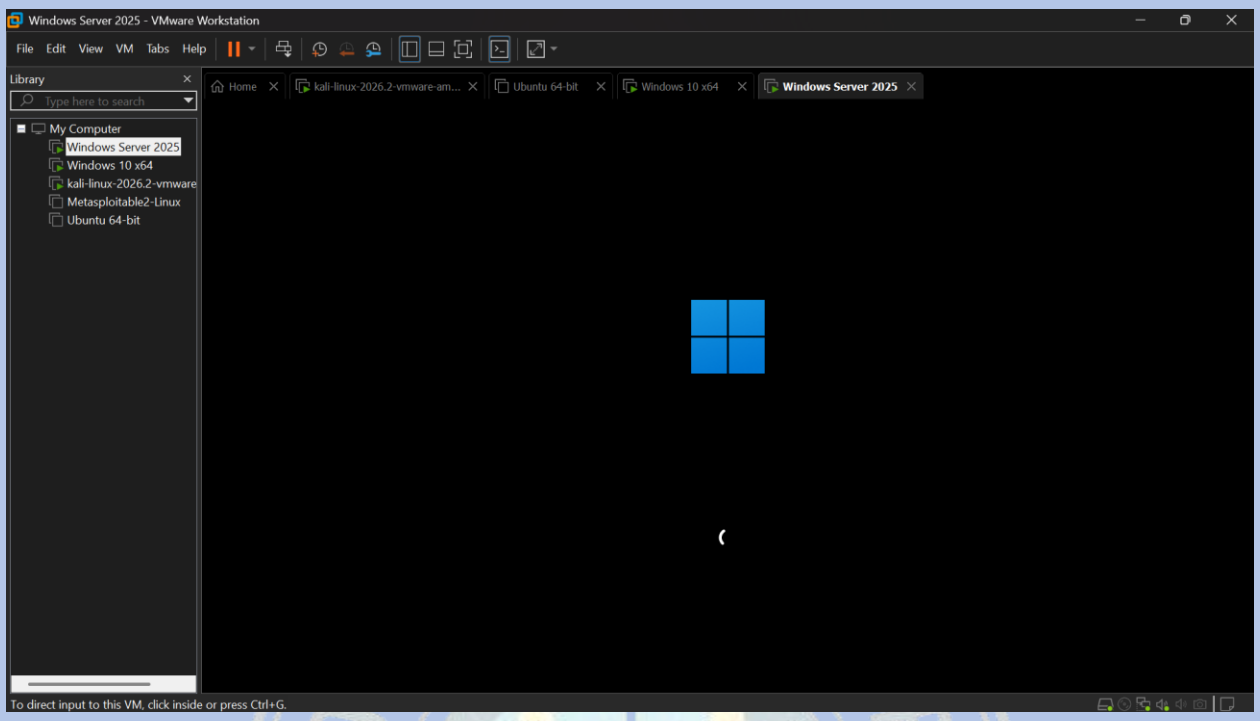


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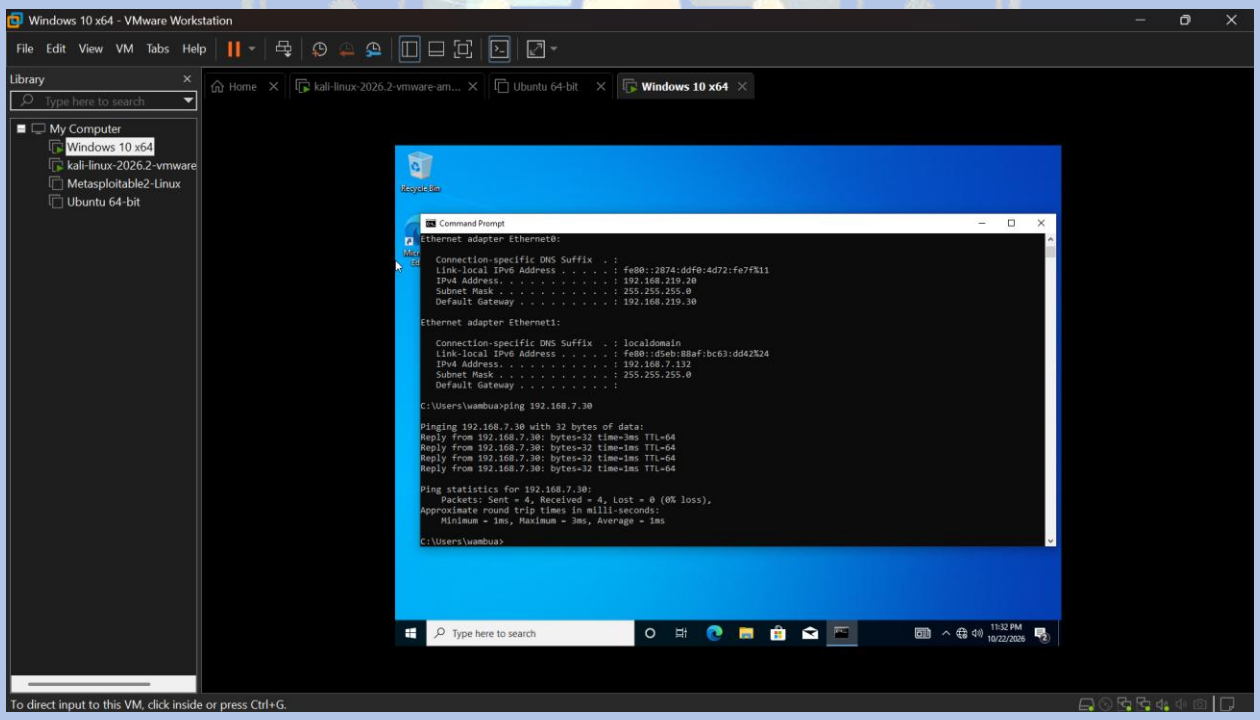


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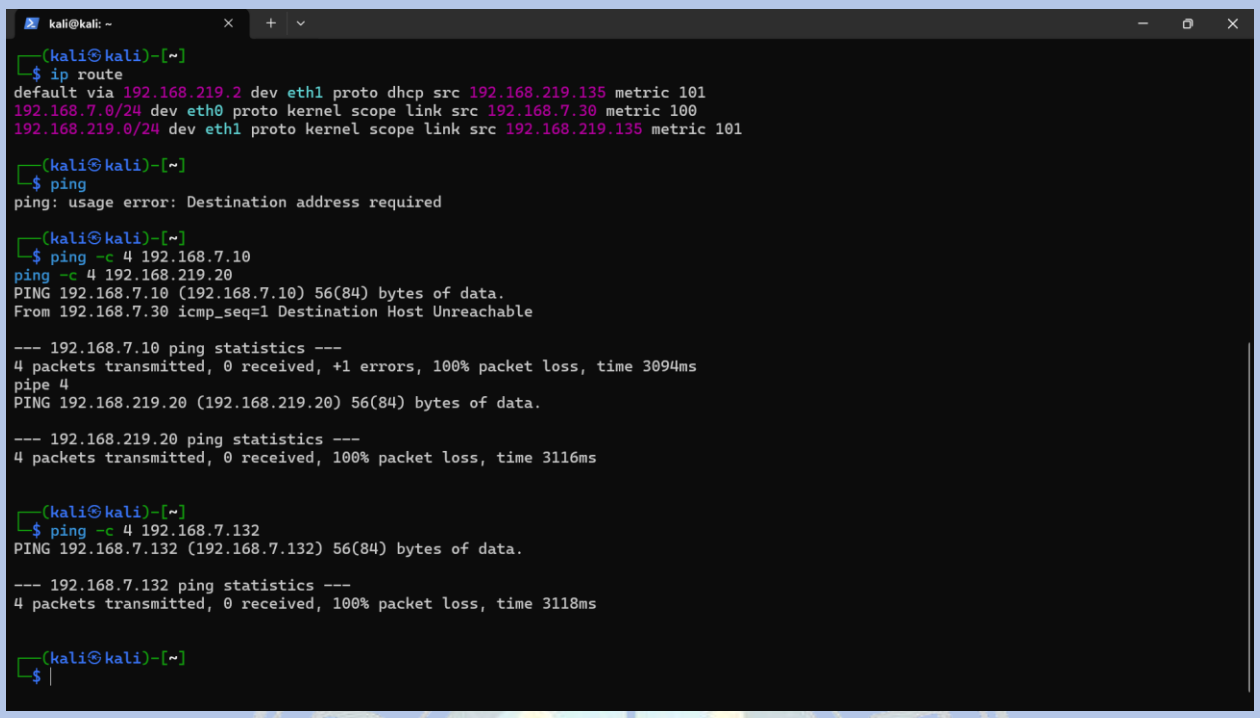


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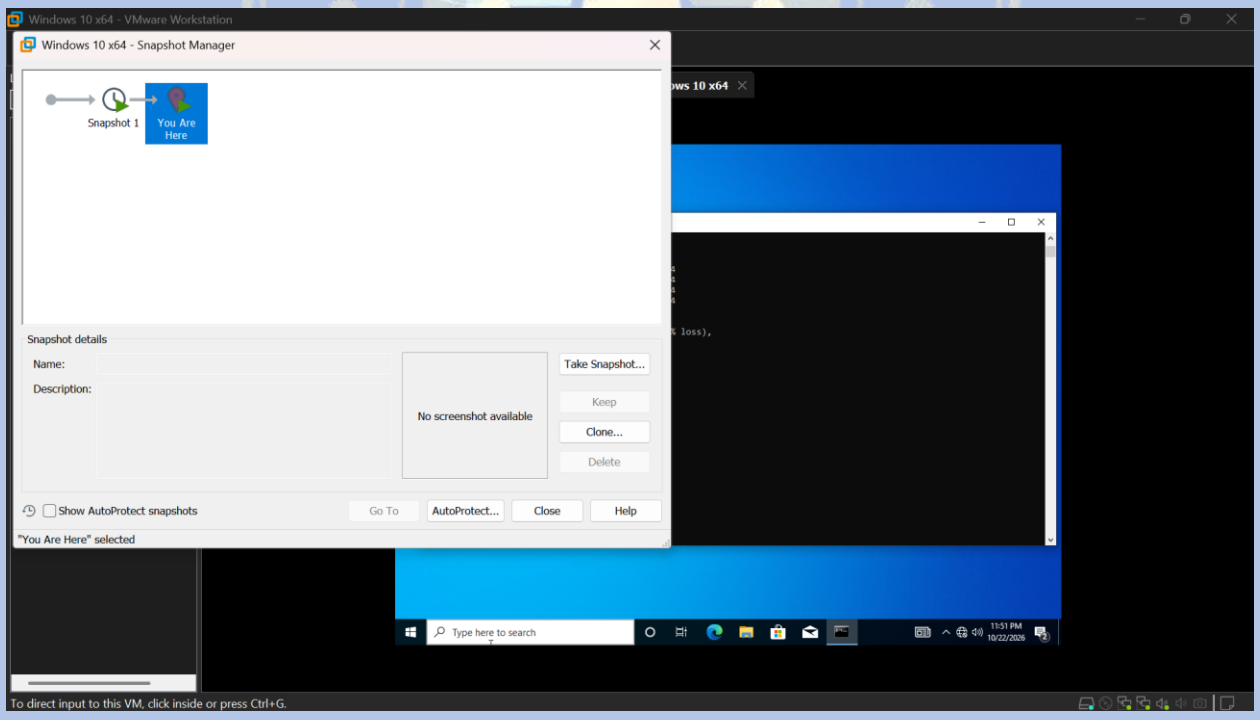


Figure 10

The purpose of the laboratory was to provide a suitable environment for practising cybersecurity concepts without unnecessarily exposing vulnerable systems to the physical network.

4.5 Linux Practical Exercises

The Linux practical exercises focused on basic system administration and monitoring.

The assigned activities included:

- Creating three Linux user accounts.
- Creating two Linux groups.
- Configuring a shared directory.
- Applying suitable directory permissions.
- Practising file-searching commands.
- Checking running processes.
- Reviewing listening network services.
- Configuring a cron job.
- Documenting Linux commands and their outputs.

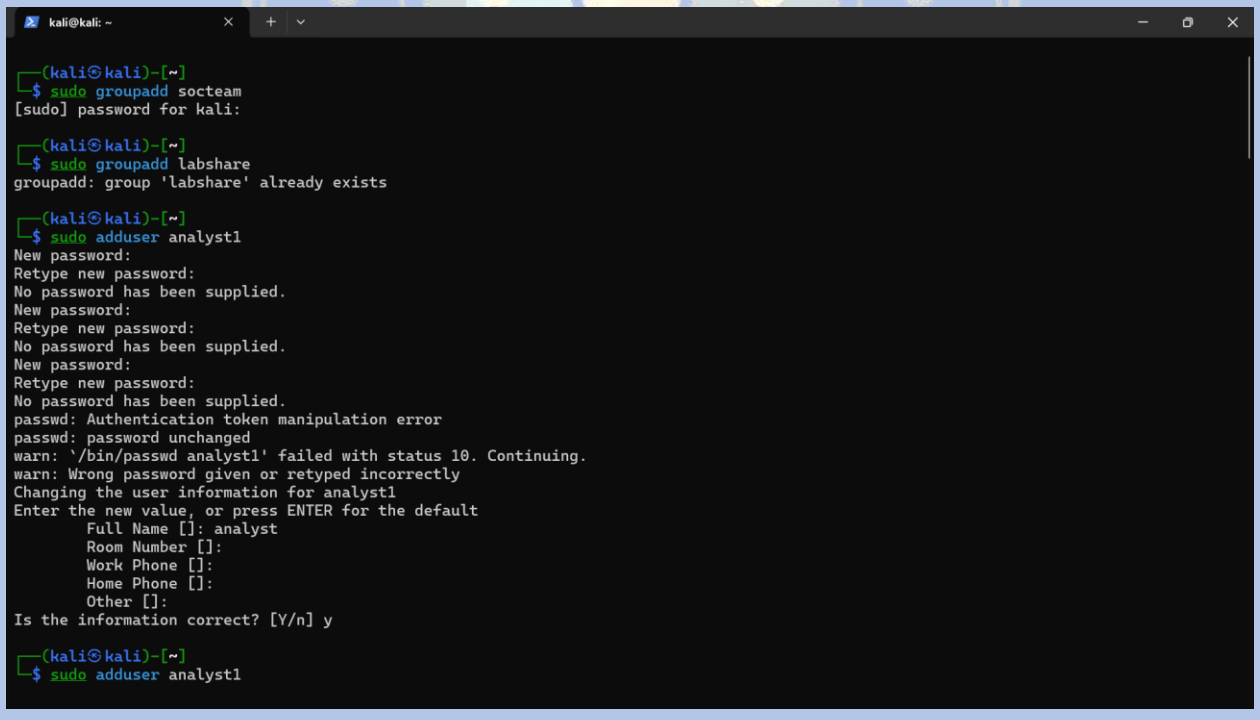


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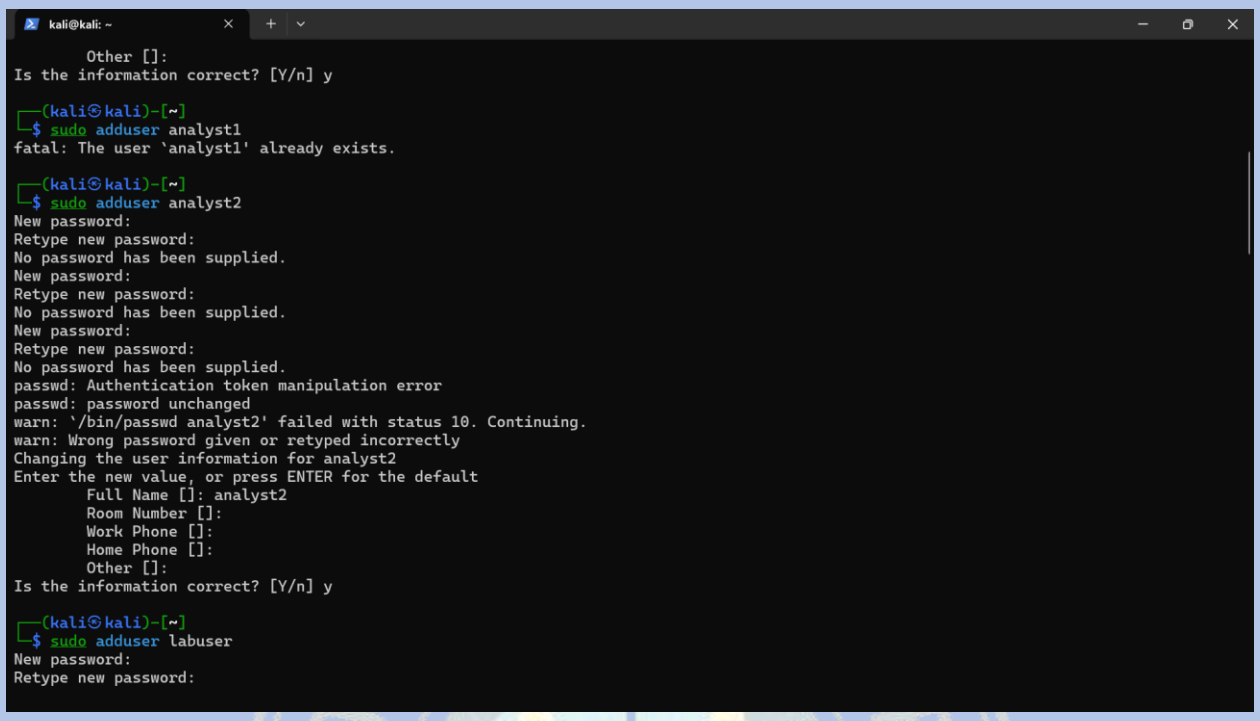


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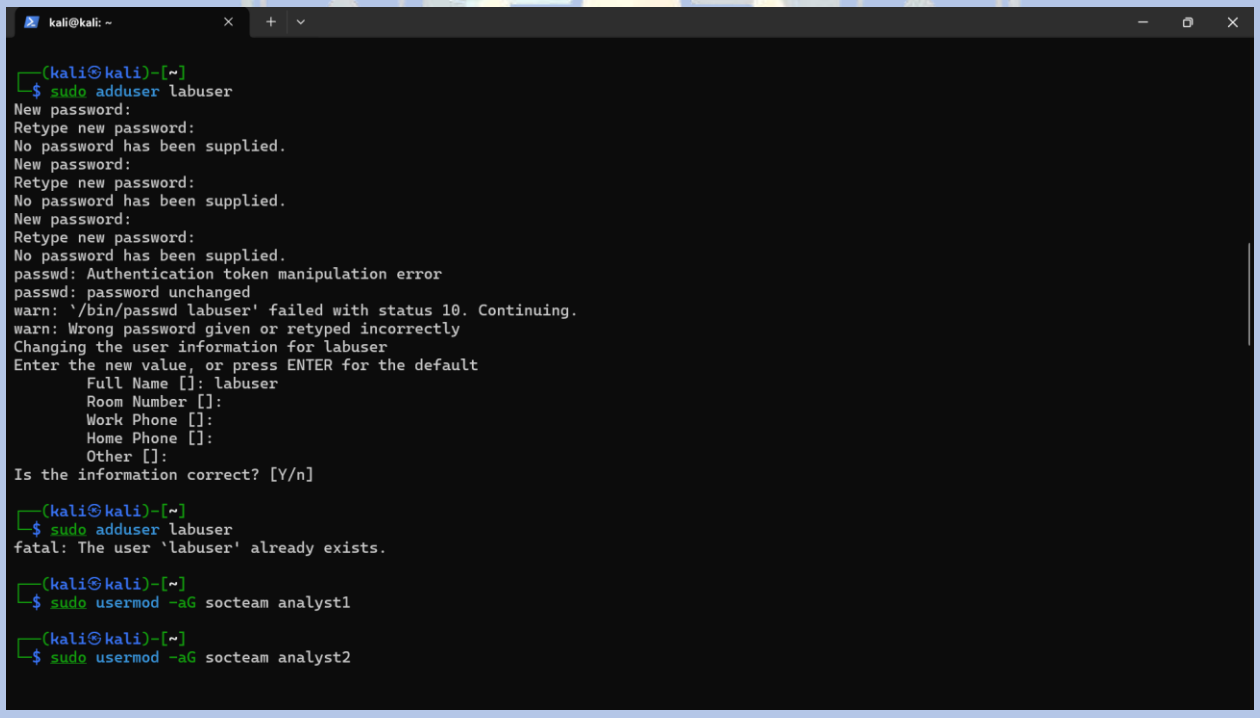


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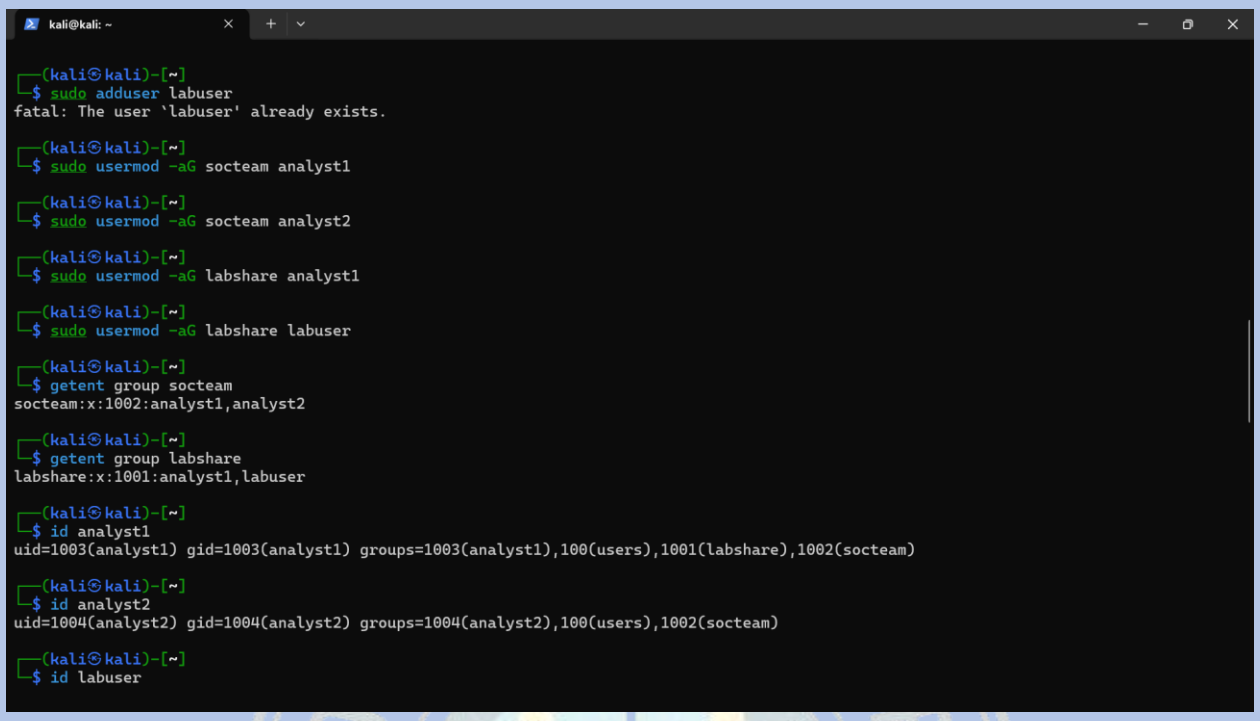


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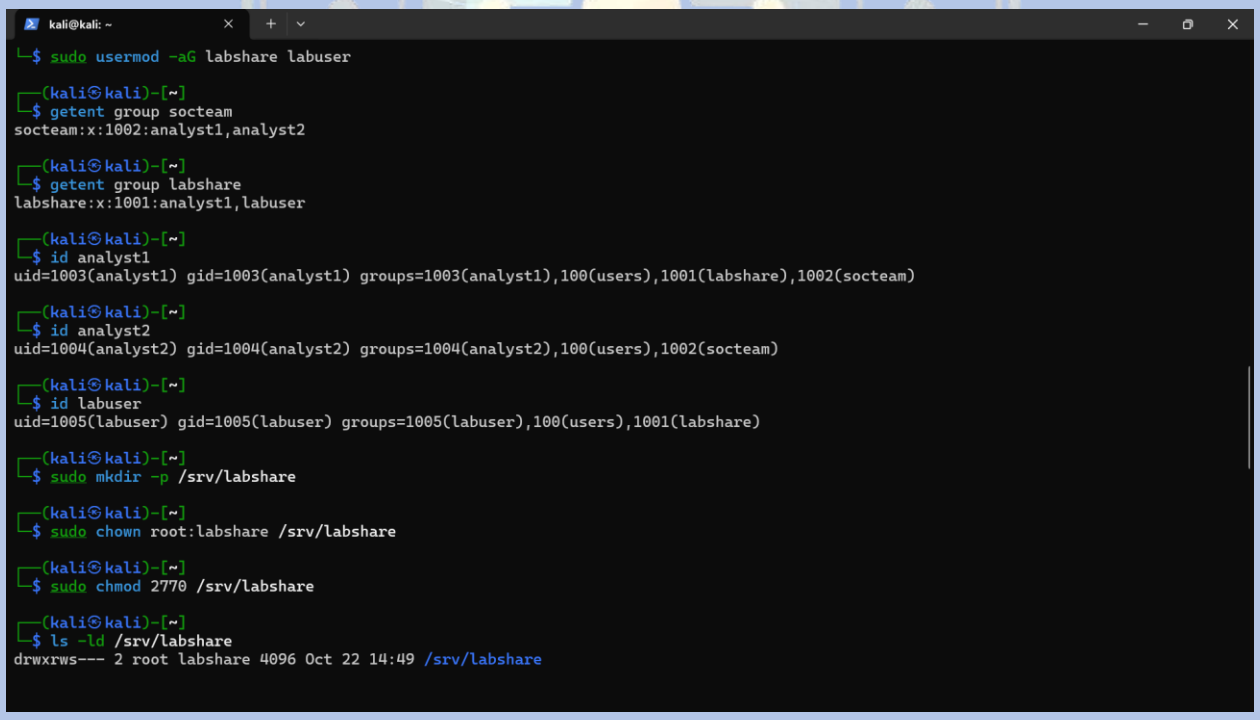


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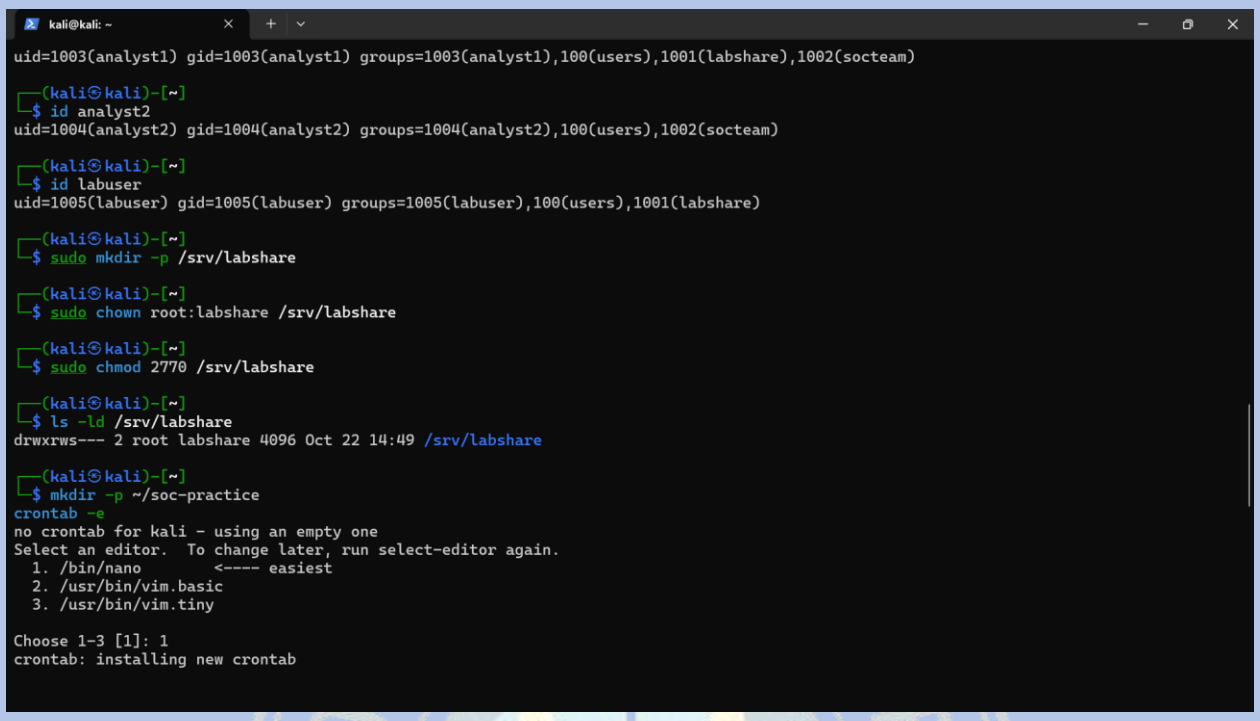


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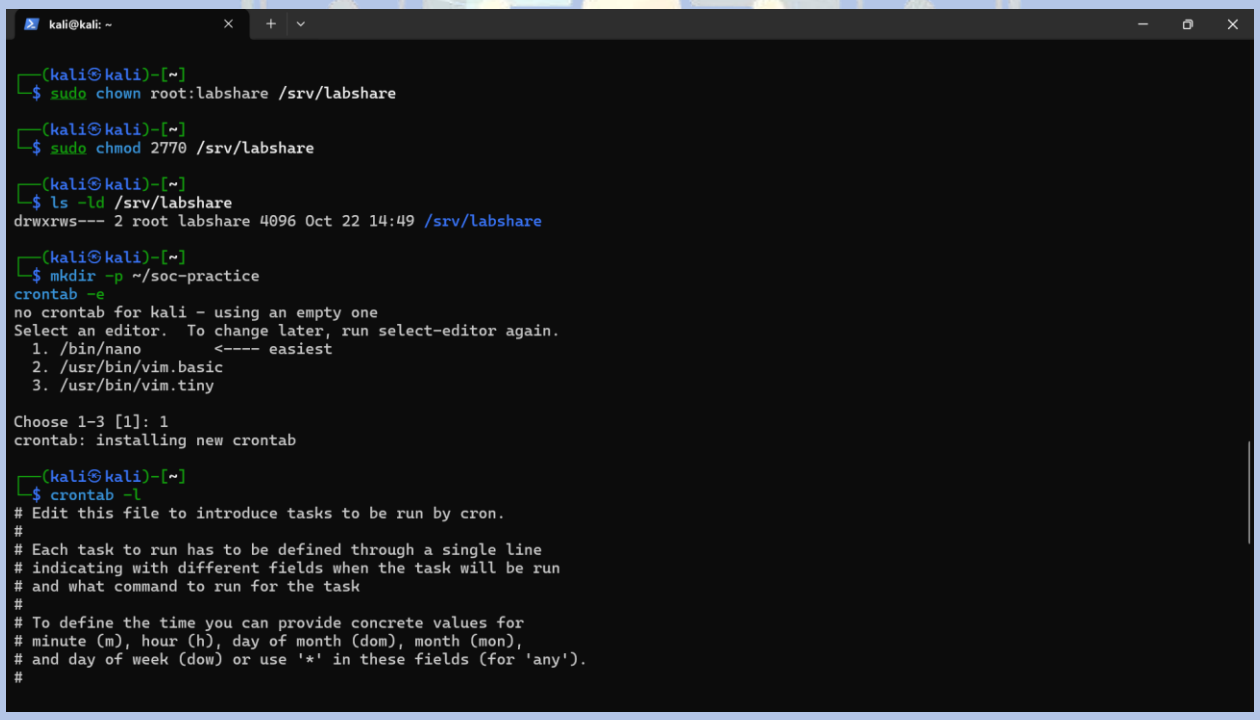


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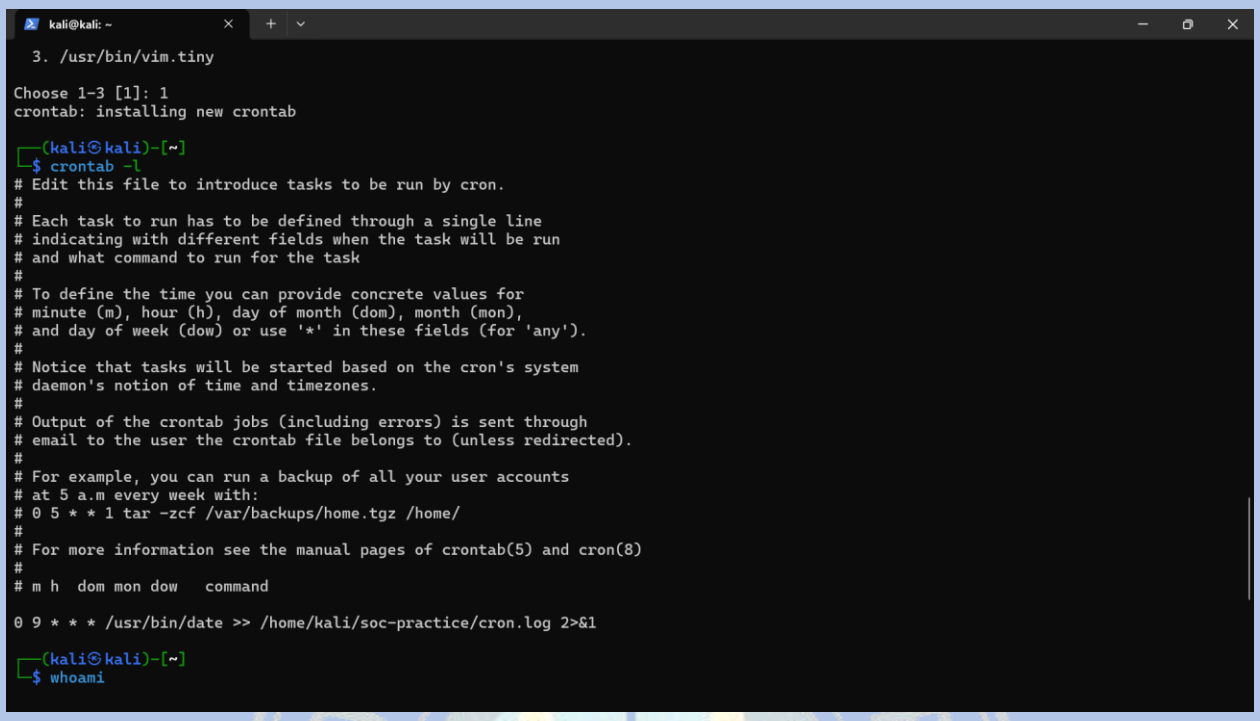


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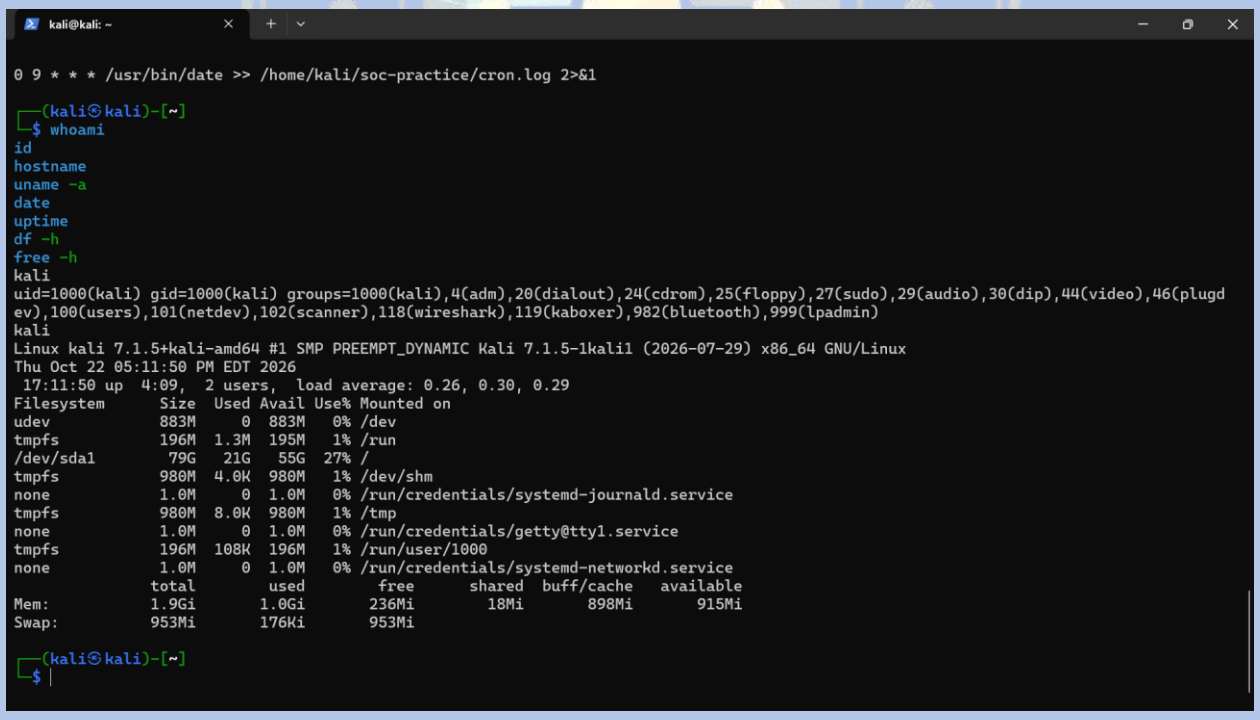


Figure 19

These exercises provided an opportunity to practise the commands and concepts studied during the theoretical sessions.

4.6 Networking Practical Exercises

The networking exercises focused on identifying network configuration information and checking connectivity between systems.

The activities included:

- Identifying IP addresses.
- Recording subnet information.
- Identifying the default gateway.
- Reviewing DNS configuration.
- Testing connectivity using ping.
- Performing DNS lookups.
- Inspecting network routes.
- Using traceroute to examine network paths.
- Preparing a network diagram.

These activities helped me understand how to collect basic network information and identify the communication paths used by computer systems.

4.7 TryHackMe Learning Activities

I worked on the assigned TryHackMe learning activities covering cybersecurity fundamentals, Linux, networking, and security concepts.

The assigned rooms were:

1. Search Skills.
2. Learning Cyber Security.
3. Linux Fundamentals Part 1.
4. What Is Networking.
5. Introductory Networking.
6. Networking Concepts.
7. Pyramid of Pain.

8. Intro to LAN.

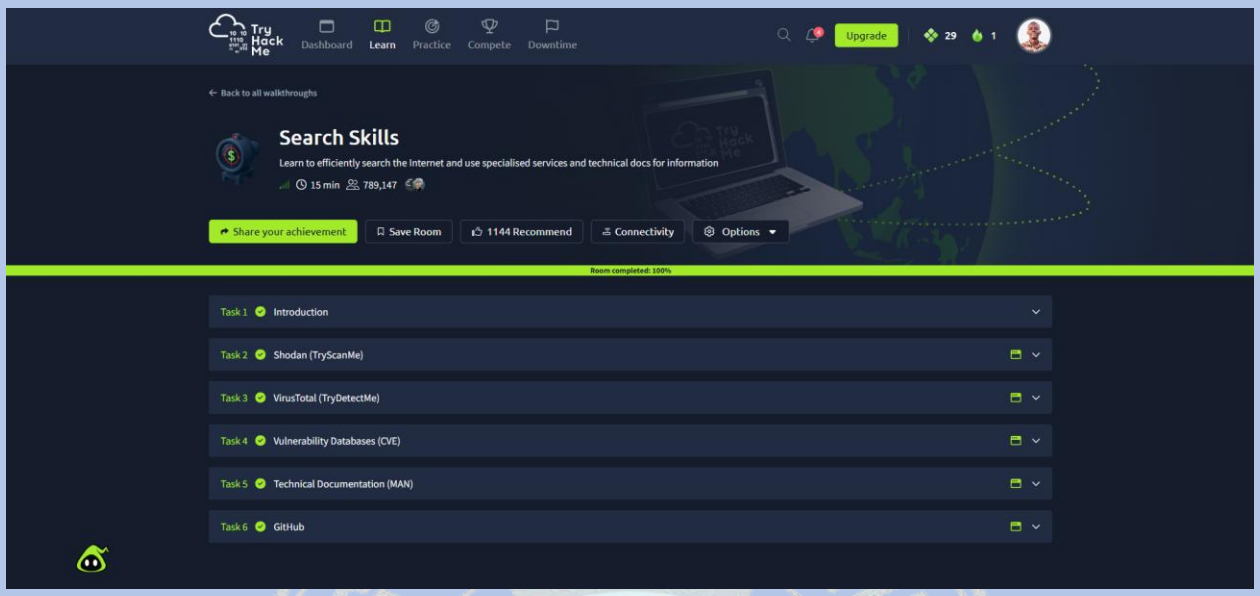


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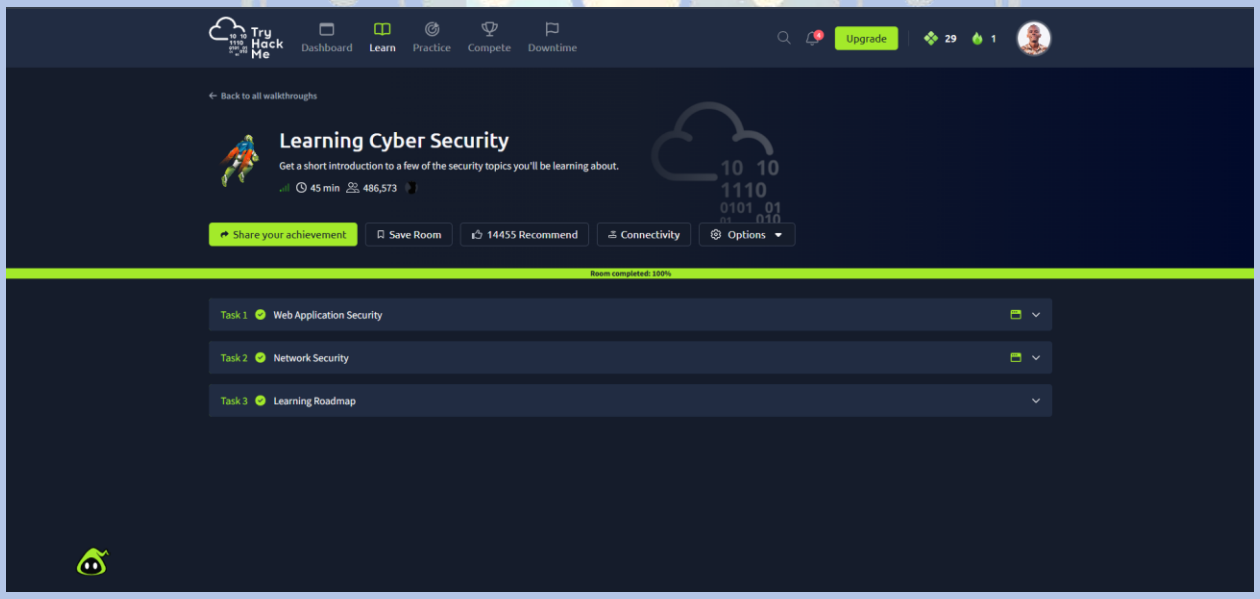


Figure 21

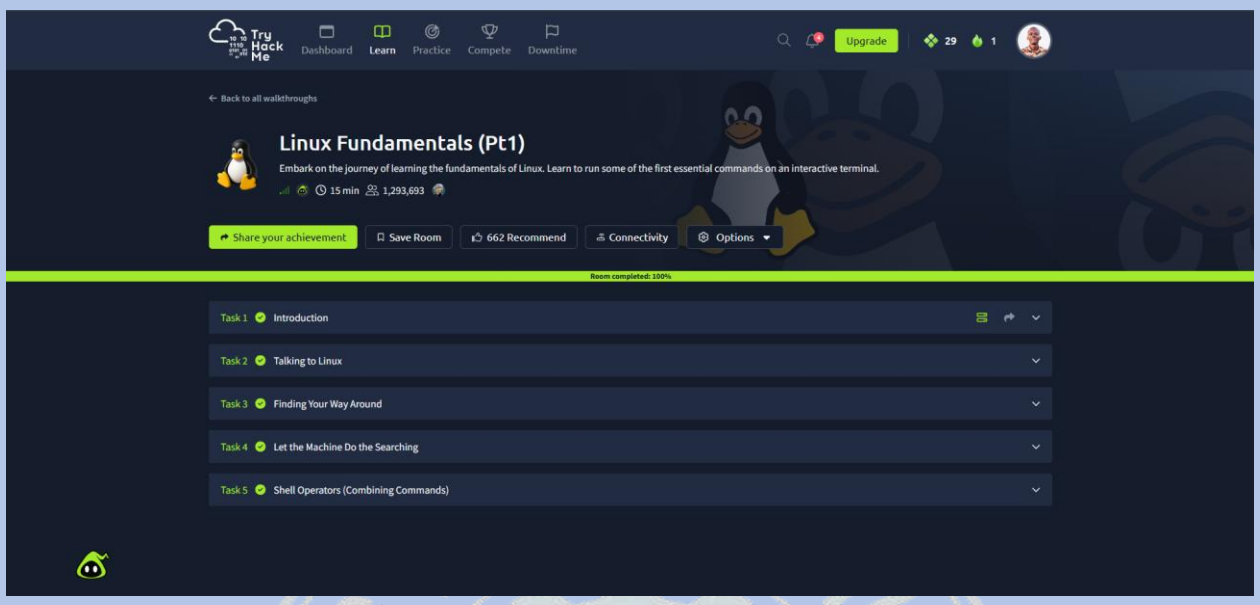


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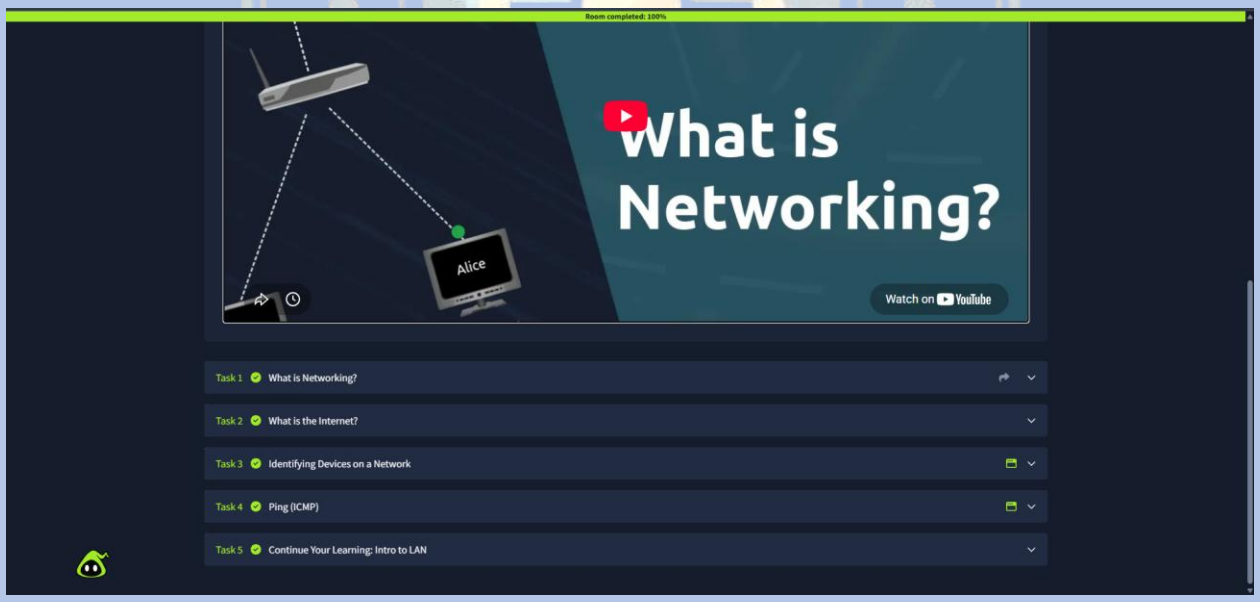


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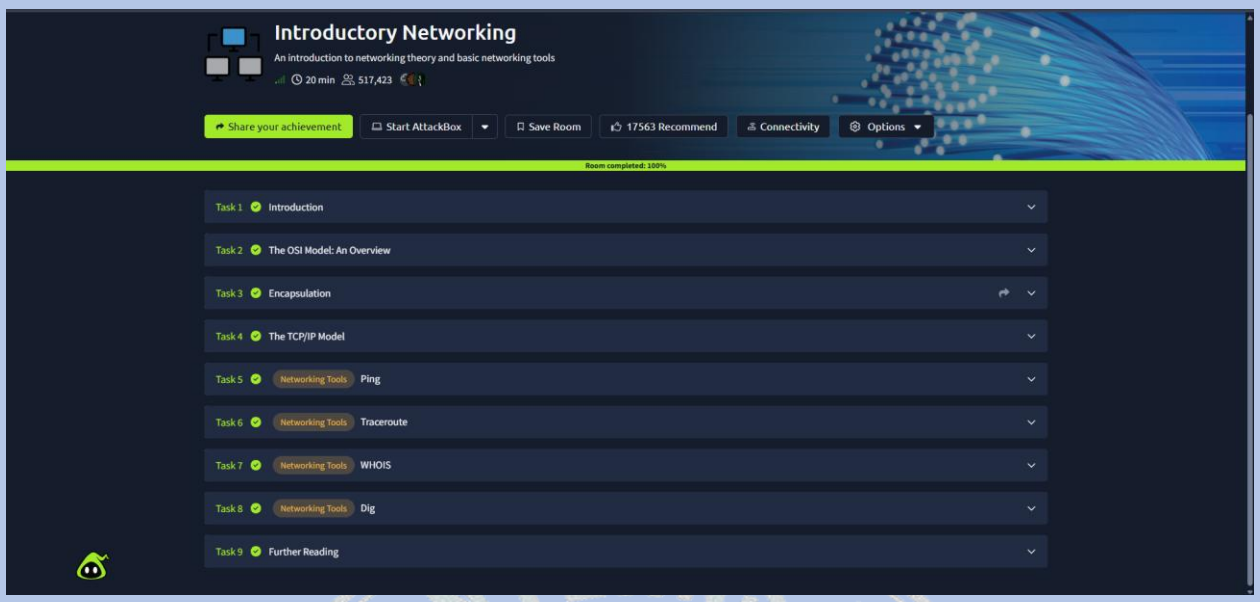


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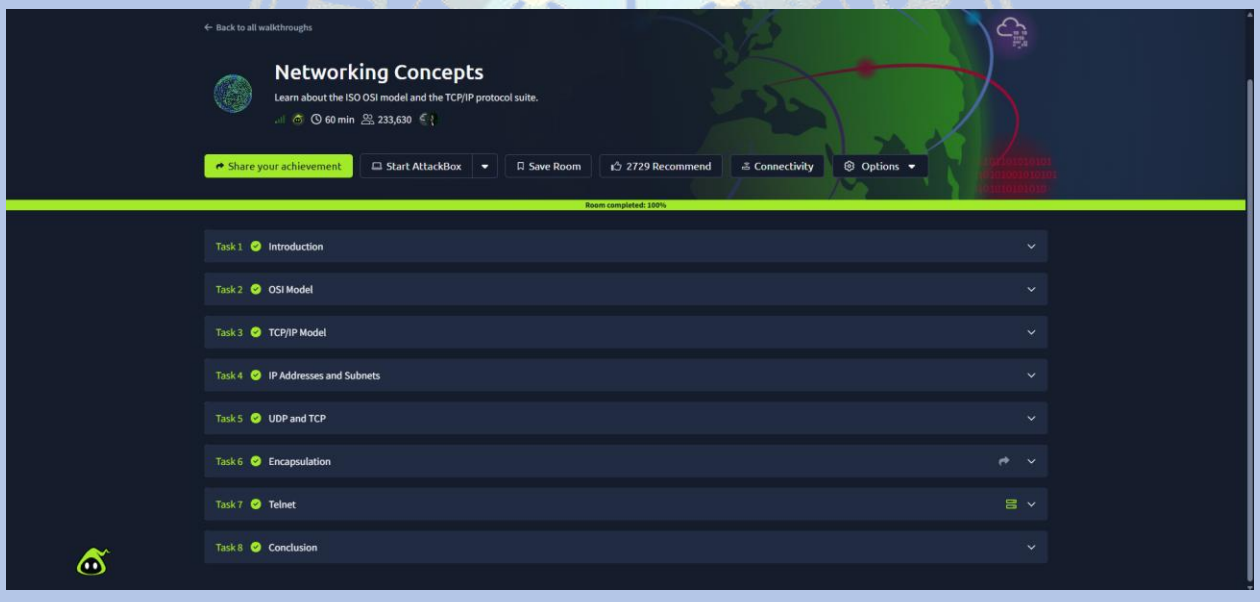


Figure 25

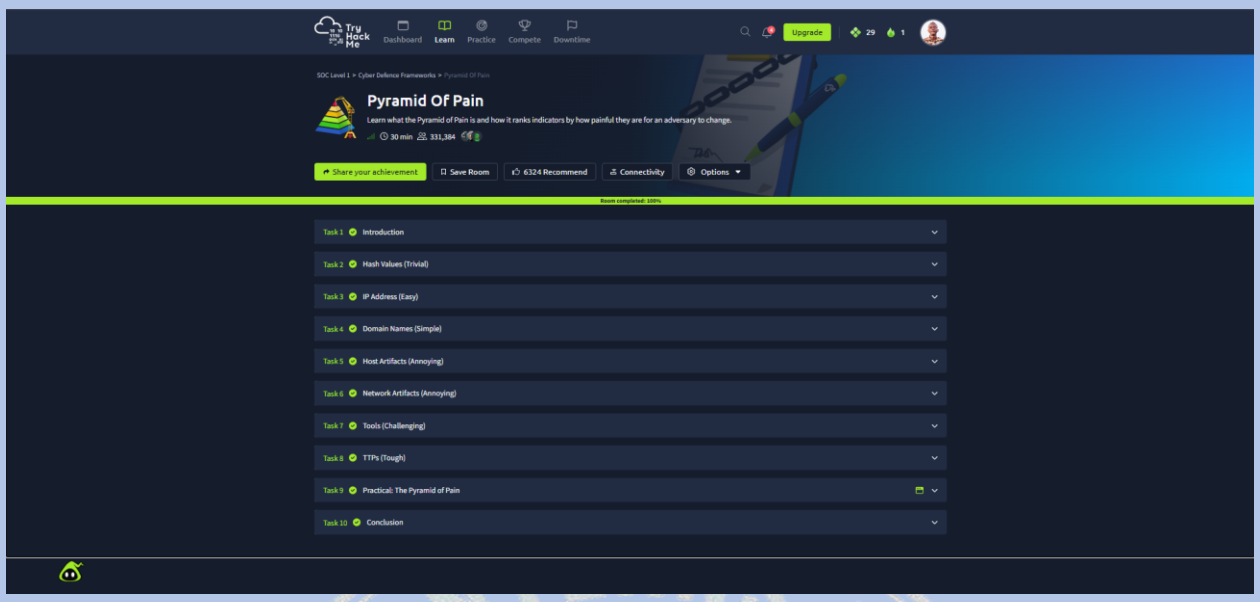


Figure 26

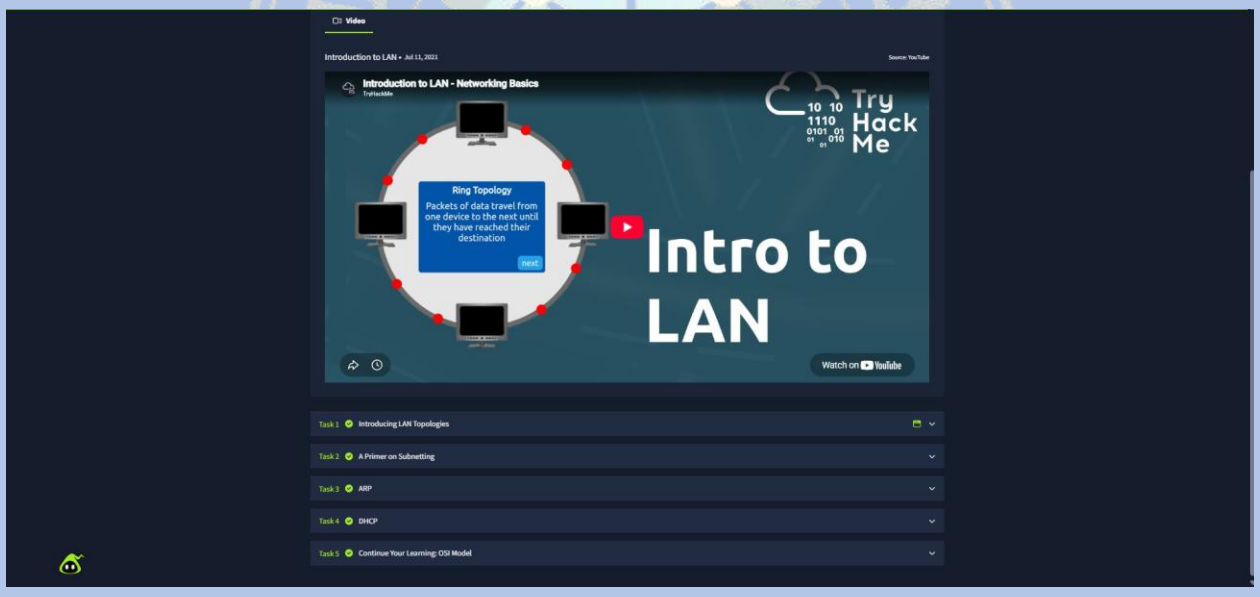


Figure 27

The activities provided guided learning opportunities to practise concepts and improve my understanding of cybersecurity.

4.8 OverTheWire Bandit

The OverTheWire Bandit exercises focused on Linux command-line practice and problem-solving.

The assigned activities covered Level 0 through Level 10. The exercises were intended to help me practise working with files, commands, and different Linux features in a controlled learning environment.

I documented the methods used and the lessons learned from the challenges. Passwords and other sensitive challenge information were excluded from the report and writeups.

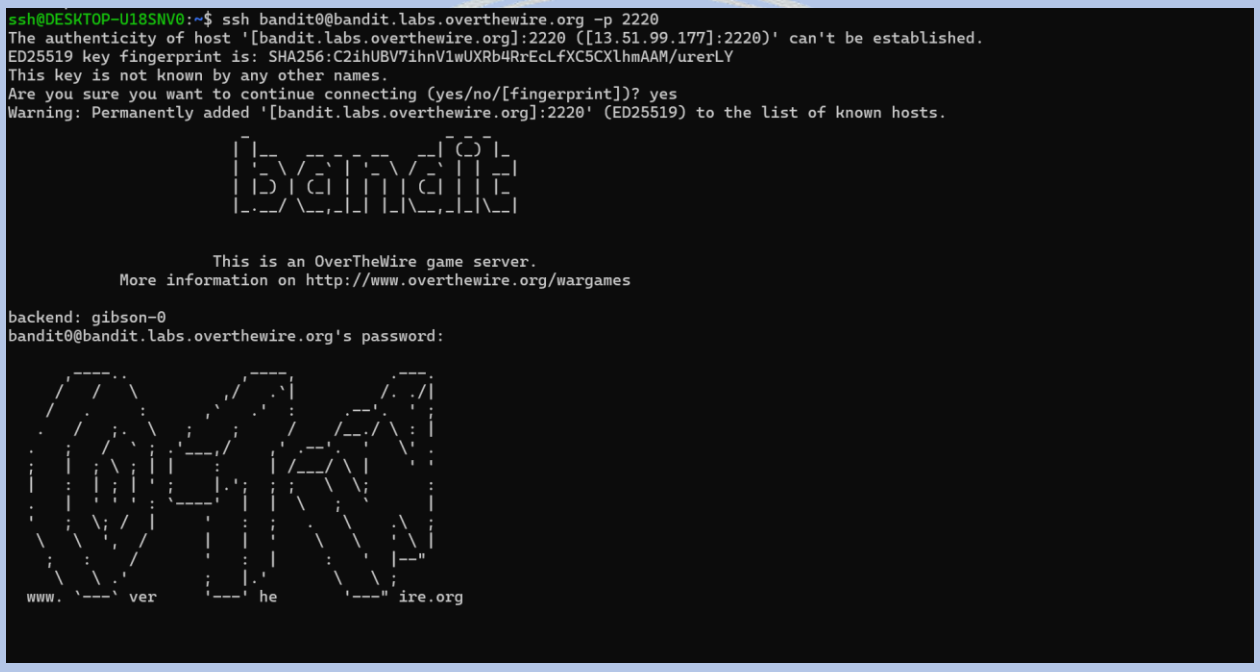


Figure 28

```
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: 6y2kwnwK6grgvwvpvLaa2T1cpFEKOhNR
```

Figure 29

```
ssh@DESKTOP-U18SNV0:~$ ssh bandit1@bandit.labs.overthewire.org -p 2220
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit1@bandit.labs.overthewire.org's password:

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.
```

Figure 30

```
bandit1@bandit:~$ cat ./-
PK8fYLZg2hnHSz83plBL1iEPKdD3QToB
bandit1@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

Figure 31

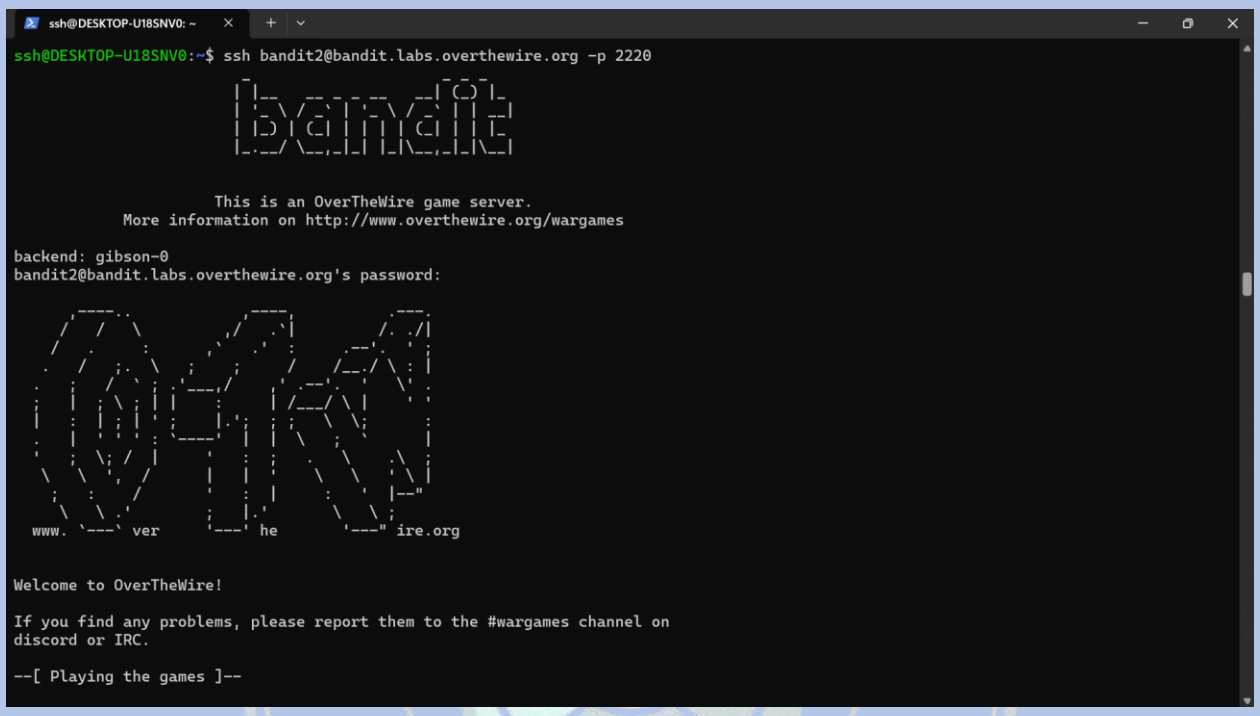


Figure 32

```
bandit2@bandit:~$ cat ./"--spaces in this filename--"
7ZZ2LFrykP2zEyvBl4m3clcL7tGYJPME
bandit2@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

Figure 33

```
ssh@DESKTOP-U18SNV0:~$ ssh bandit3@bandit.labs.overthewire.org -p 2220
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit3@bandit.labs.overthewire.org's password:

      ,----..            ,----,          .---.
     /   /   \         ,/   .`|         /. ./|
    /   .     :      ,`   .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .--'.  '   \' .
  ;   |  ; \ ; | |    :     | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;   \  \;      :
  .   |  ' ' ' : `----'  |  |  \   ;  `      |
  '   ;  \; /  |      '   :  ;   .   \    .\  ;
   \   \  ',  /       |   |  '    \   \   ' \ |
    ;   :    /        '   :  |     :   '  |--"
     \   \ .'         ;   |.'       \   \ ;
  www. `---` ver      '---' he        '---" ire.org


Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.
```

Figure 34

```
bandit3@bandit:~$ cd inhere
ls -la
total 12
drwxr-xr-x 2 root    root    4096 Sep 26 21:54 .
drwxr-xr-x 3 root    root    4096 Sep 26 21:54 ..
-rw-r----- 1 bandit4 bandit3   33 Sep 26 21:54 ...Hiding-From-You
bandit3@bandit:~/inhere$ cat .hidden
cat: .hidden: No such file or directory
bandit3@bandit:~/inhere$ cat Hiding-From-You
cat: Hiding-From-You: No such file or directory
bandit3@bandit:~/inhere$ cat ...Hiding-From-You
xzTXq1rDJQVVAzdv5cHq1TQytTWufAMq
bandit3@bandit:~/inhere$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

Figure 35

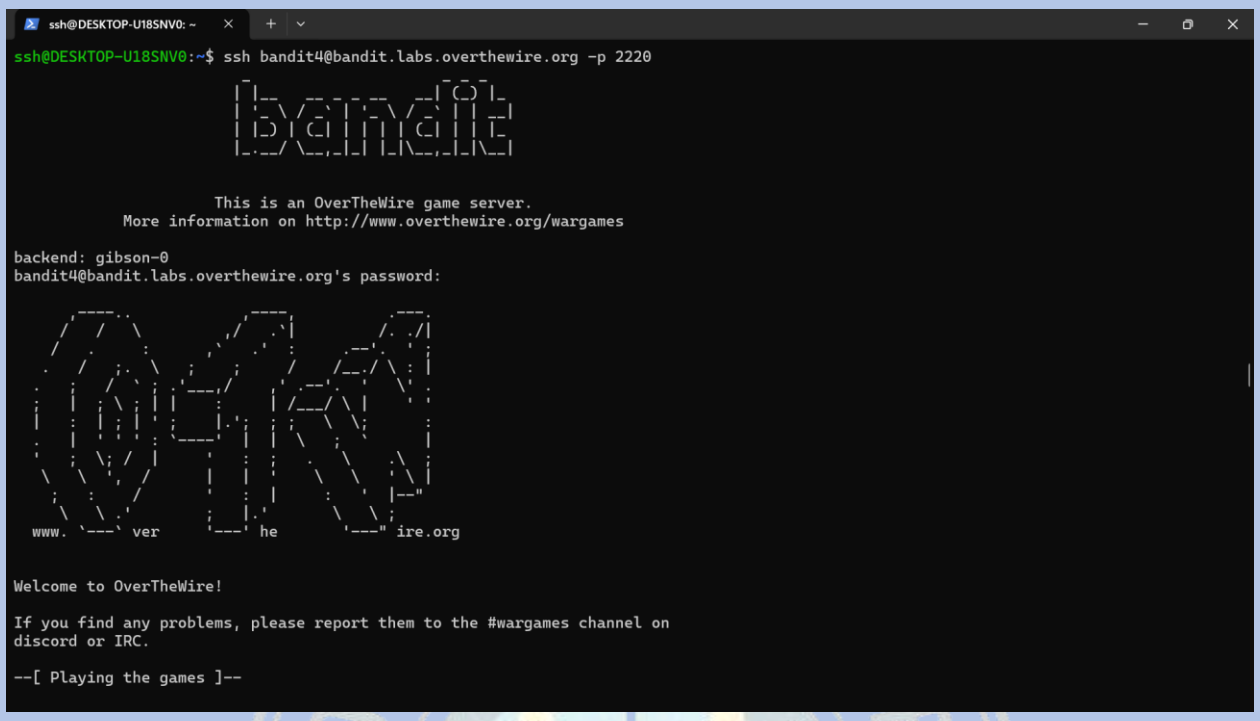


Figure 36

```
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ dir
-file00  -file01  -file02  -file03  -file04  -file05  -file06  -file07  -file08  -file09
bandit4@bandit:~/inhere$ file ./*
./-file00: data
./-file01: data
./-file02: data
./-file03: data
./-file04: data
./-file05: data
./-file06: data
./-file07: ASCII text
./-file08: data
./-file09: data
bandit4@bandit:~/inhere$ cat ./-file07
6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG
bandit4@bandit:~/inhere$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

Figure 37

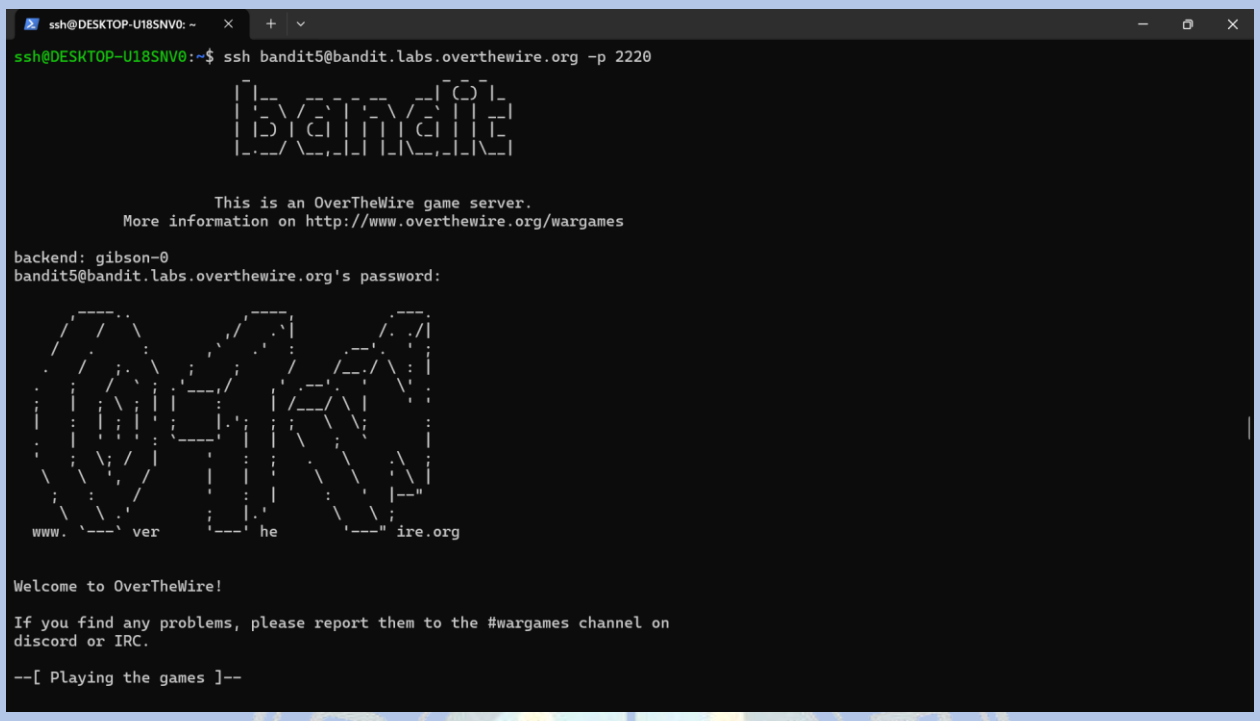


Figure 38

```
bandit5@bandit:~$ find inhere -type f -size 1033c ! -executable
inhere/maybehere07/.file2
bandit5@bandit:~$ cat inhere/maybehere07/.file2
pXa26xhMWaC2SvDotA4r9EgZkulOeSBW
```

Figure 39

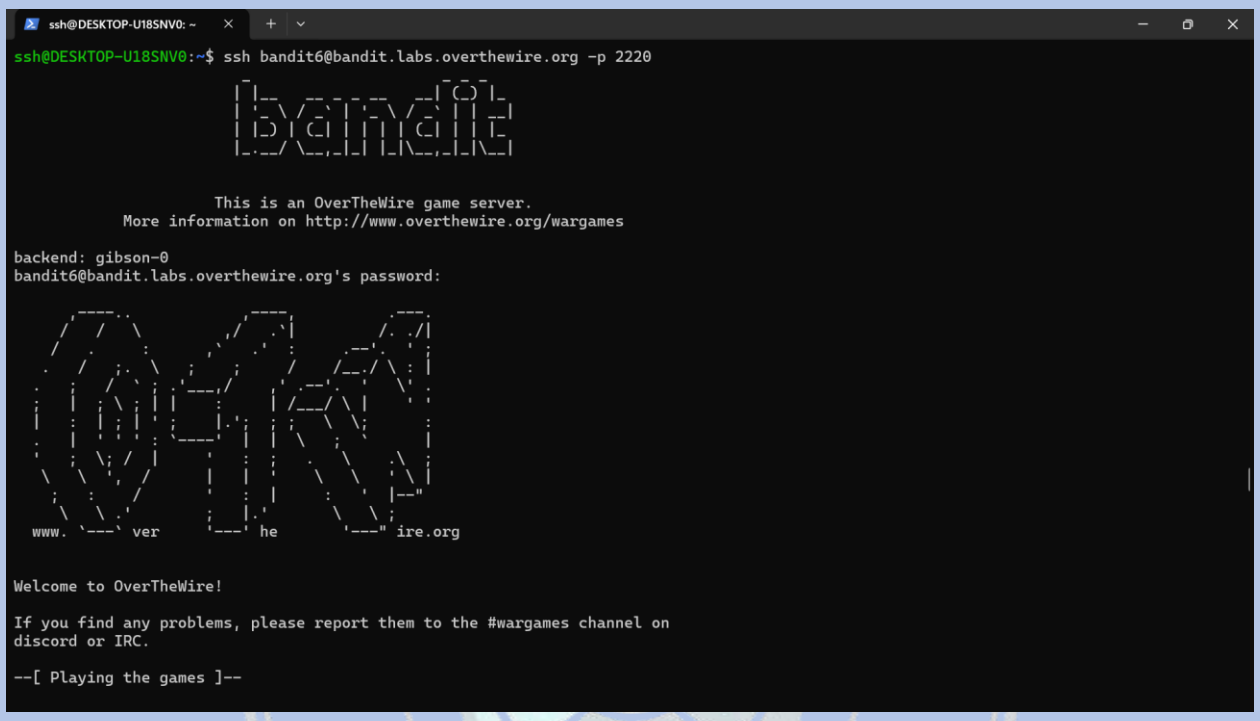


Figure 40

```
bandit6@bandit:~$ find / -user bandit7 -group bandit6 -size 33c 2>/dev/null
/var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
Bmnnvf82KzQlfxgAI2d1zYbr1u9pr3E3
bandit6@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

Figure 41

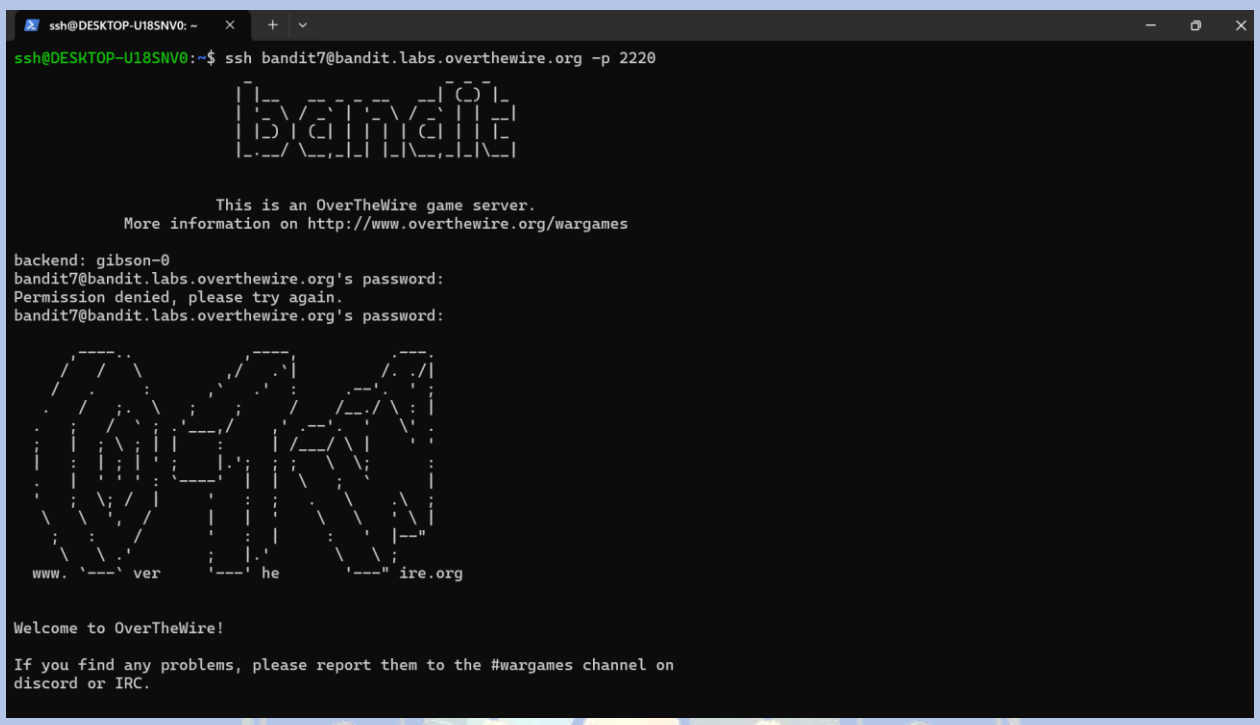


Figure 42

```
bandit7@bandit:~$ grep "millionth" data.txt
millionth       VR1ljMayciFxbnUokuQmJFw6QC9VKtub
bandit7@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

Figure 43

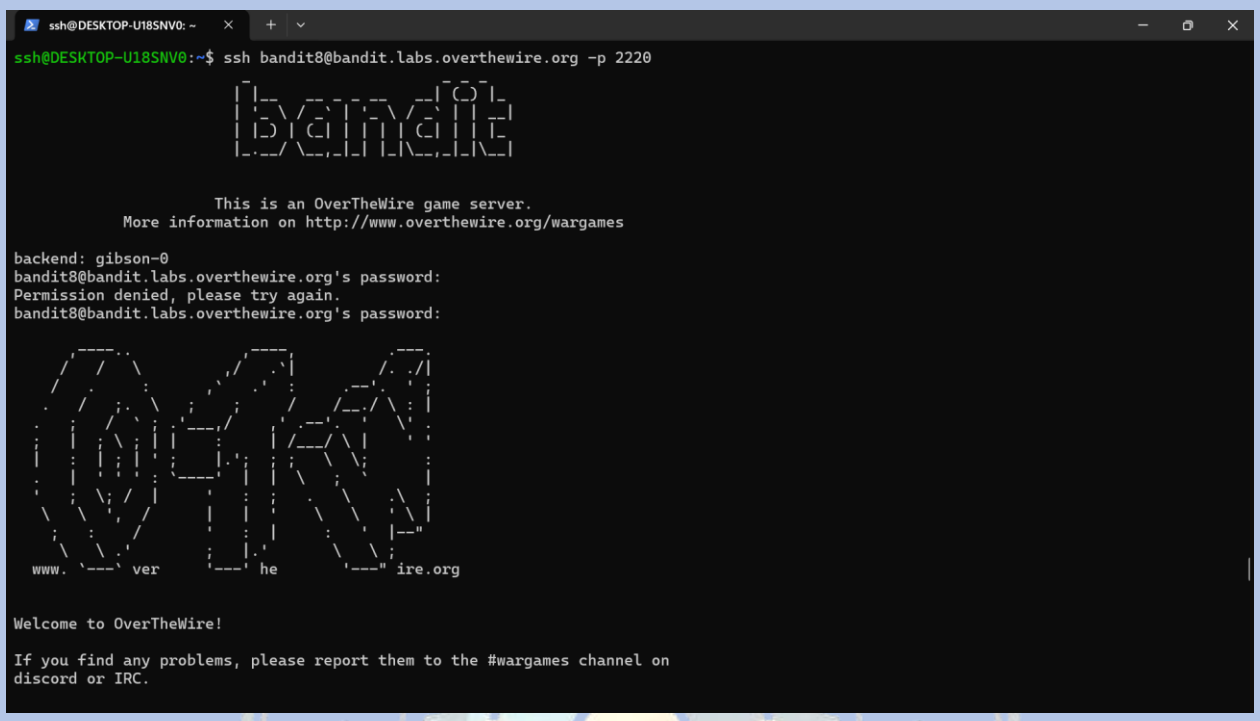


Figure 44

```
bandit8@bandit:~$ sort data.txt | uniq -u
EjmOSvuAu7sGAHqHVcBDPirRe9T03kxl
bandit8@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

Figure 45

```
ssh@DESKTOP-U18SNV0:~$ ssh bandit9@bandit.labs.overthewire.org -p 2220
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit9@bandit.labs.overthewire.org's password:
Permission denied, please try again.
bandit9@bandit.labs.overthewire.org's password:

      ,----..            ,----,         .---.
     /   /   \       ,//   .`|        /. ./|
    /   .     :     ,`//    .`:    .--'.  ' ;
   .   /   ;.  \  ;  ;     /  /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .--'.  '  \ .
  ;   |  ; \ ; | |   :     | /___/ \ |    ' '
  |   :  | ; | ' ;   |.';  ; ;   \  \;      :
  .   |  ' ' ' : `----'  |  |  \   ;  `      |
  '   ;  \; /  |      '  :  ;   .\  \    .\  ;
   \   \  ',  /       |  |  '    \  \   ' \ |
    ;   :    /        '  :  |     :   '  |--"
     \   \ .'         ;  |.'       \   \ ;
  www. `---` ver      '---' he       '---" ire.org


Welcome to OverTheWire!
```

Figure 46

```
bandit9@bandit:~$ strings data.txt | grep "="
\========== the
q:=V
K        *=
=zZw
w=-1
vc.=
========== password
<u*========== is
=9$s
{=S[
;a =
lp>IoY-b==ld
========== B0s2khmbT9u0geKuOoVGW3JZKhndE3BG
ec=}
         =ulD
```

Figure 47

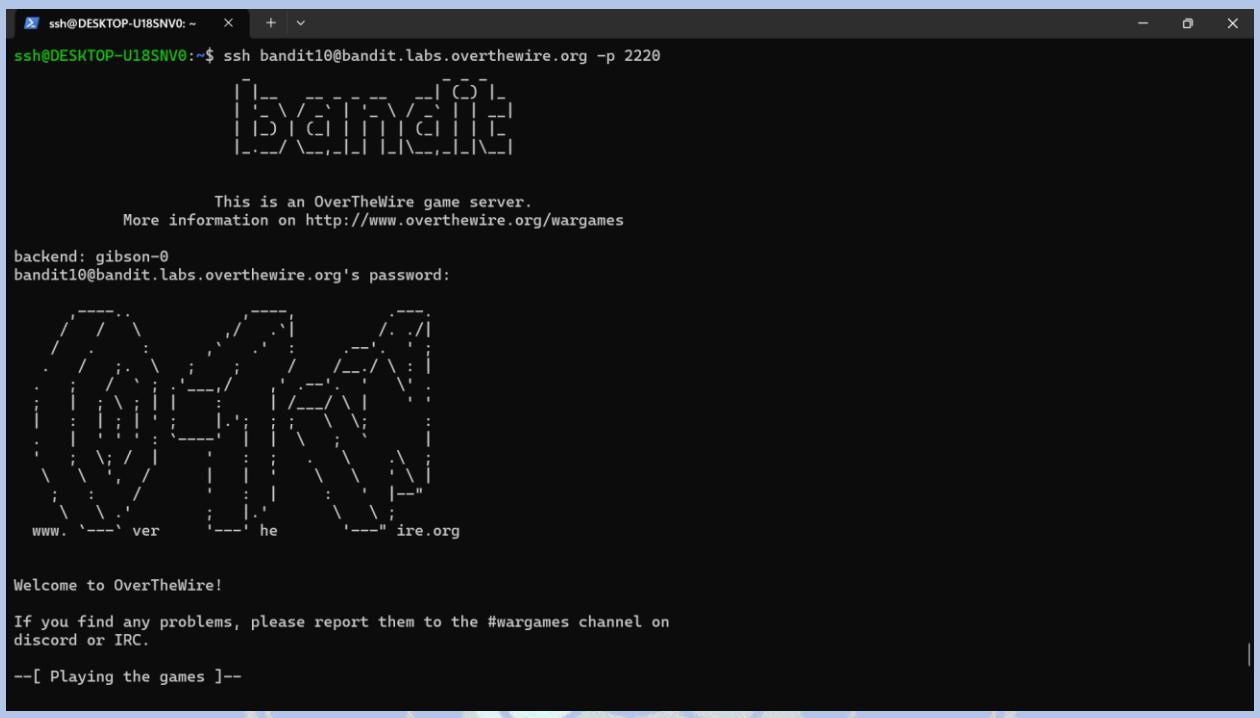


Figure 48

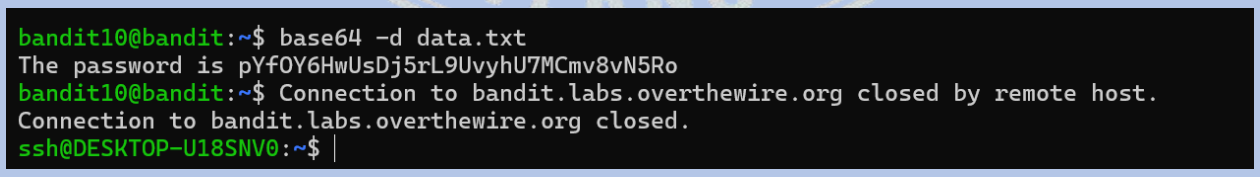


Figure 49

Learning outcome: I developed more familiarity with Linux commands, file inspection, and solving technical challenges step by step.

5. FINDINGS AND LESSONS LEARNED

Week 1 helped me understand how cybersecurity, networking, and operating system administration are connected.

Through the theoretical activities, I learned why confidentiality, integrity, and availability are important when protecting information. I also gained a clearer understanding of the different responsibilities of SOC analysts, penetration testers, and incident responders.

The networking topics helped me understand how IP addresses, DNS, DHCP, ports, and protocols support communication between computer systems. This knowledge is useful when investigating network activity and identifying services that may require monitoring.

The Linux exercises introduced me to user management, groups, file permissions, process monitoring, and basic system administration. I also learned that properly managing permissions helps prevent unauthorized access to files and directories.

The virtual laboratory activities highlighted the importance of keeping cybersecurity practice within a controlled environment. Using an isolated network and taking virtual machine snapshots can help reduce risks and make it easier to restore a system after an exercise.

The guided learning activities also helped me apply the concepts studied during the week.

Key Learning Outcomes

By the end of Week 1, I had:

1. Improved my understanding of cybersecurity fundamentals.
2. Learned the importance of the CIA triad.
3. Developed a clearer understanding of SOC analyst responsibilities.
4. Learned how penetration testing and incident response differ.
5. Improved my knowledge of networking models and protocols.
6. Practised Linux command-line operations.
7. Learned more about Linux user and group management.
8. Improved my understanding of file permissions.
9. Practised identifying network configuration information and testing connectivity.
10. Understood the importance of authorization and laboratory isolation during cybersecurity activities.

6. CHALLENGES AND RESOLUTIONS

During the activities completed, I did not encounter any major technical challenges that prevented me from continuing with the assigned learning tasks.

The exercises helped me become more familiar with the instructions, tools, and laboratory requirements.

Where further practice was needed, I reviewed the relevant learning materials and command documentation.

7. EVIDENCE AND DELIVERABLES

The Week 1 work is organized into folders to make it easier to review the learning activities and supporting materials.

Folder	Contents
Notes	Cybersecurity theory, networking, Linux, and legal and ethical notes
Writeups	Individual TryHackMe room and Bandit level-transition writeups
Findings	Screenshots and supporting practical evidence
Lab Setup	Virtual machine information, network configuration, and connectivity evidence
Linux Practice	User accounts, groups, permissions, commands, and cron documentation
Network Map	Network inventory and laboratory diagram
OverTheWire Bandit	Challenge writeups and completion documentation
Cheat Sheet	Linux command reference
Reports	Weekly report and presentation

The practical screenshots are stored separately in the Findings folder. They are not repeated or attached in this report.

This report provides a summary of the activities and learning outcomes, while the Findings folder contains the supporting practical evidence.

8. PROFESSIONAL AND ETHICAL CONSIDERATIONS

During the cybersecurity learning activities, I considered the importance of responsible and authorized testing.

The exercises were intended for controlled educational environments. Cybersecurity tools and techniques should only be used within an approved scope and with the necessary authorization.

The following principles guided the activities:

- Perform security testing only with appropriate authorization.
- Respect the approved scope of activities.
- Protect passwords, credentials, and confidential information.
- Avoid exposing vulnerable laboratory systems to unnecessary networks.
- Document findings accurately.
- Maintain the confidentiality and integrity of collected evidence.
- Follow applicable Kenyan cybersecurity and data protection requirements.

The Computer Misuse and Cybercrimes Act, 2018, as amended, and the Data Protection Act, 2019, are important legal considerations when carrying out cybersecurity activities in Kenya.

9. CONCLUSION

Week 1 provided a useful foundation for my cybersecurity internship. The activities introduced me to important concepts in cybersecurity, Linux administration, networking, and virtual laboratory preparation.

I gained a better understanding of how computer systems communicate, how Linux systems are managed, and how basic security principles help protect digital information.

The practical exercises also helped me become more familiar with Linux commands, network configuration, connectivity testing, and guided cybersecurity learning platforms.

One of the main lessons from the week was the importance of practising cybersecurity in a controlled and authorized environment. I also learned that accurate documentation and proper evidence management are important parts of technical work.

The knowledge and skills gained during Week 1 will support my continued learning in networking, Linux administration, security monitoring, and incident investigation.

10. REFERENCES

1. Kenya Law – Computer Misuse and Cybercrimes Act, 2018, as amended.
2. Office of the Data Protection Commissioner, Kenya.
3. Cisco Networking Academy.
4. TryHackMe – Cybersecurity Learning Platform.
5. OverTheWire – Bandit.
6. Oracle VirtualBox Documentation.
7. Linux Manual Pages

DECLARATION

I confirm that this report is a summary of my Week 1 cybersecurity internship learning activities. Practical completion details and results should reflect the work I actually performed.